

SMART HR OPERATIONS AND WORKFORCE MANAGEMENT SYSTEM

Software Design and Architecture

Deliverable 3

Group Number # 16

Ms. Laiba Imran

Employee Management	Leave & Attendance	Performance & Compliance
Sohaib(24i-3108) UC-01 to UC-05	Shaiman(24i-3074) UC-06 to UC-10	Hasaam(24i-3107) UC-11 to UC-15

Table of Contents

Table of Contents	2
1. Introduction.....	4
2. System Sequence Diagrams (SSDs)	5
2.1 Employee Management Module (UC-01 – UC-05).....	5
UC-01 — Register New Employee.....	5
UC-02 — Update Employee Record	8
UC-03 — Assign Employee to Department	10
UC-04 — Initiate Employee Offboarding	12
UC-05 — View Employee Profile.....	14
2.2 Leave and Attendance Management Module (UC-06 – UC-10)	16
UC-06 — Submit Leave Request	16
UC-07 — Approve or Reject Leave Request	18
UC-08 — Record Daily Attendance	20
UC-09 — Request Attendance Correction	22
UC-10 — View Leave Balance and Attendance Summary.....	24
2.3 Performance and Compliance Management Module (UC-11 – UC-15)	26
UC-11 — Initiate Performance Evaluation Cycle	26
UC-12 — Submit Employee Performance Evaluation	28
UC-13 — Monitor Employee Probation Status	30
UC-14 — Generate Compliance Report	32
UC-15 — Generate HR Analytics Report	34
3. Sequence Diagrams (SDs)	36
3.1 Employee Management Module (UC-01 – UC-05).....	36
UC-01 – Sequence Diagrams.....	36
UC-02 – Sequence Diagrams.....	37
UC-03 – Sequence Diagrams.....	38
UC-04 – Sequence Diagrams.....	39
UC-05 – Sequence Diagrams.....	40
3.2 Leave and Attendance Management Module (UC-06 – UC-10)	41
UC-06 – Sequence Diagrams.....	41
UC-07 – Sequence Diagrams.....	42
UC-08 – Sequence Diagrams.....	43
UC-09 – Sequence Diagrams.....	44

UC-10 – Sequence Diagrams.....	45
3.3 Performance and Compliance Management Module (UC-11 – UC-15)	46
UC-11 – Sequence Diagrams.....	46
UC-12 – Sequence Diagrams.....	47
UC-13 – Sequence Diagrams.....	48
UC-14 – Sequence Diagrams.....	49
UC-15 – Sequence Diagrams.....	50
4. Class Diagram	51
4.1 Class Diagram	51
5. Refined Domain Model.....	52

1. Introduction

This document is the third deliverable for the Smart HR Operations and Workforce Management System project. Building upon the use case specifications and domain model established in Deliverable 2, this deliverable focuses on the dynamic and structural modelling of the system through System Sequence Diagrams (SSDs), Sequence Diagrams (SDs), and a Class Diagram (CD).

The document has been structured so that each group member's allocated use cases (UC-01 to UC-05 for Employee Management, UC-06 to UC-10 for Leave & Attendance, and UC-11 to UC-15 for Performance & Compliance) are presented in dedicated sections. Each use case includes:

- System Sequence Diagrams (SSDs) for the main success scenario and major alternative flows.
- Sequence Diagrams (SDs) for each system event/operation identified, annotated with GRASP pattern notes.

The final section presents the consolidated Class Diagram, which integrates all operations from the SDs and reflects GRASP and Gang of Four (GoF) design pattern assignments.

2. System Sequence Diagrams (SSDs)

This section presents System Sequence Diagrams for all fifteen use cases. For each use case a Main Success Scenario SSD is provided, followed by SSDs for the three major alternative / exception flows.

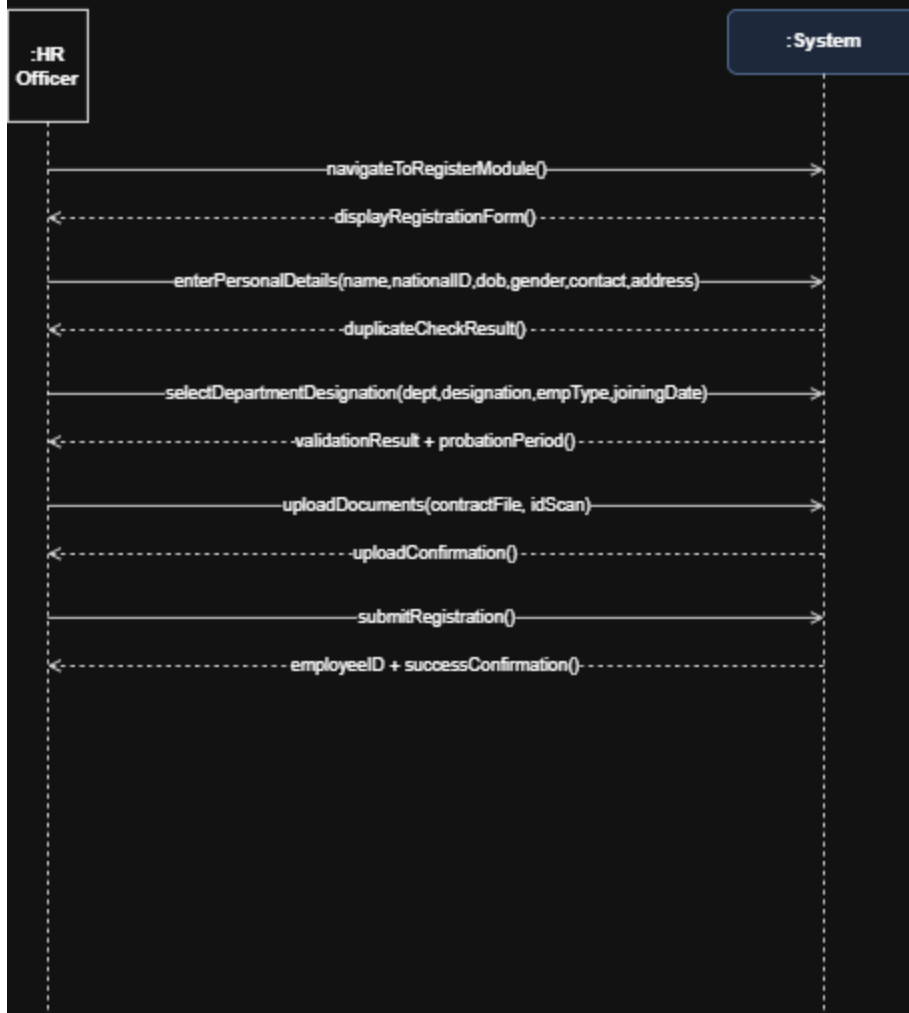
2.1 Employee Management Module (UC-01 – UC-05)

UC-01 — Register New Employee

Use Case ID	UC-01
Use Case Name	Register New Employee
Primary Actor	HR Officer
Description	HR Officer registers a new employee by entering personal details, department, designation, and employment type. The system validates all fields, generates a unique Employee ID, persists the record, and dispatches a welcome notification.

Main Success Scenario SSD:

UC-01: Register New Employee — SSD (Main Success Scenario)



Major Alternative / Exception Flow SSDs:



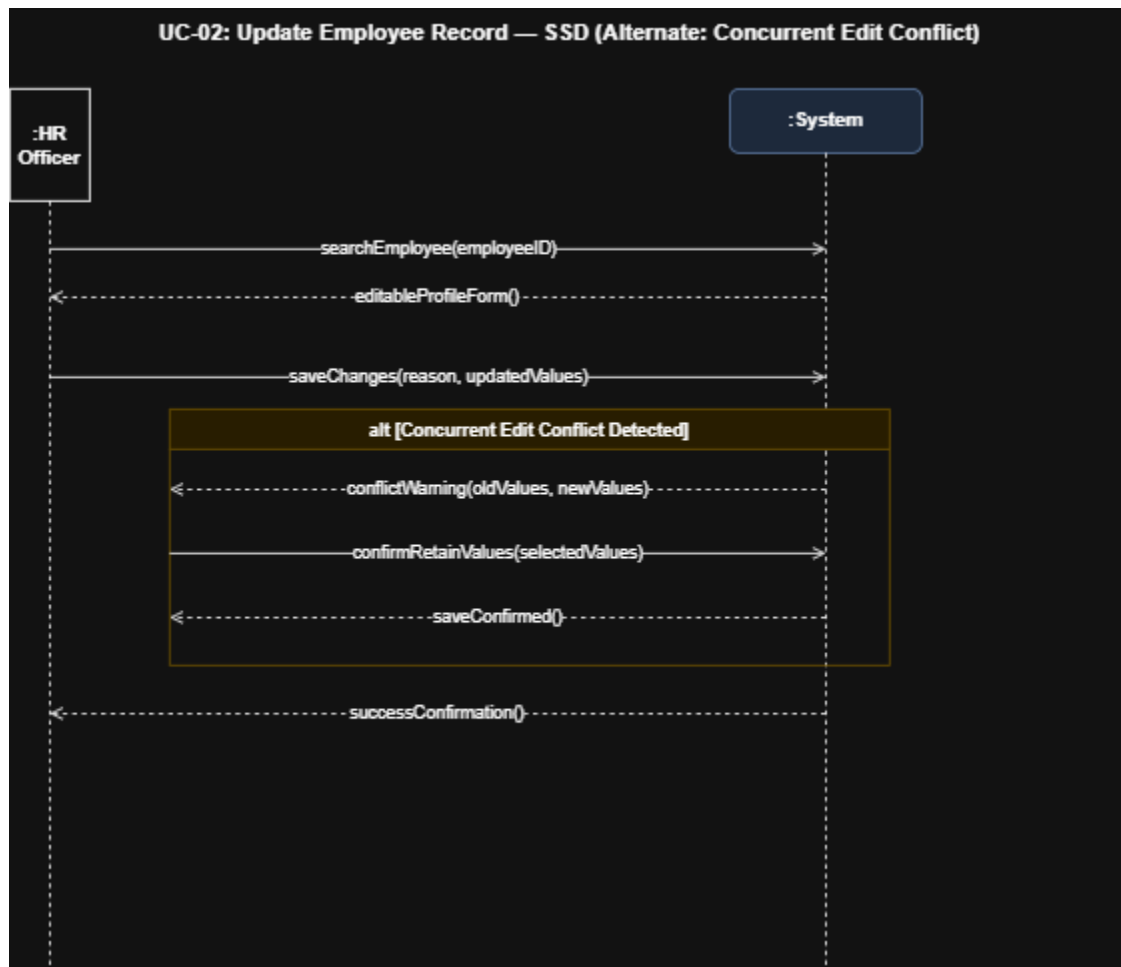
UC-02 — Update Employee Record

Use Case ID	UC-02
Use Case Name	Update Employee Record
Primary Actor	HR Officer
Description	HR Officer locates an employee record and modifies one or more fields. System validates changes, writes a timestamped audit log capturing old and new values, and saves the updated record.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:



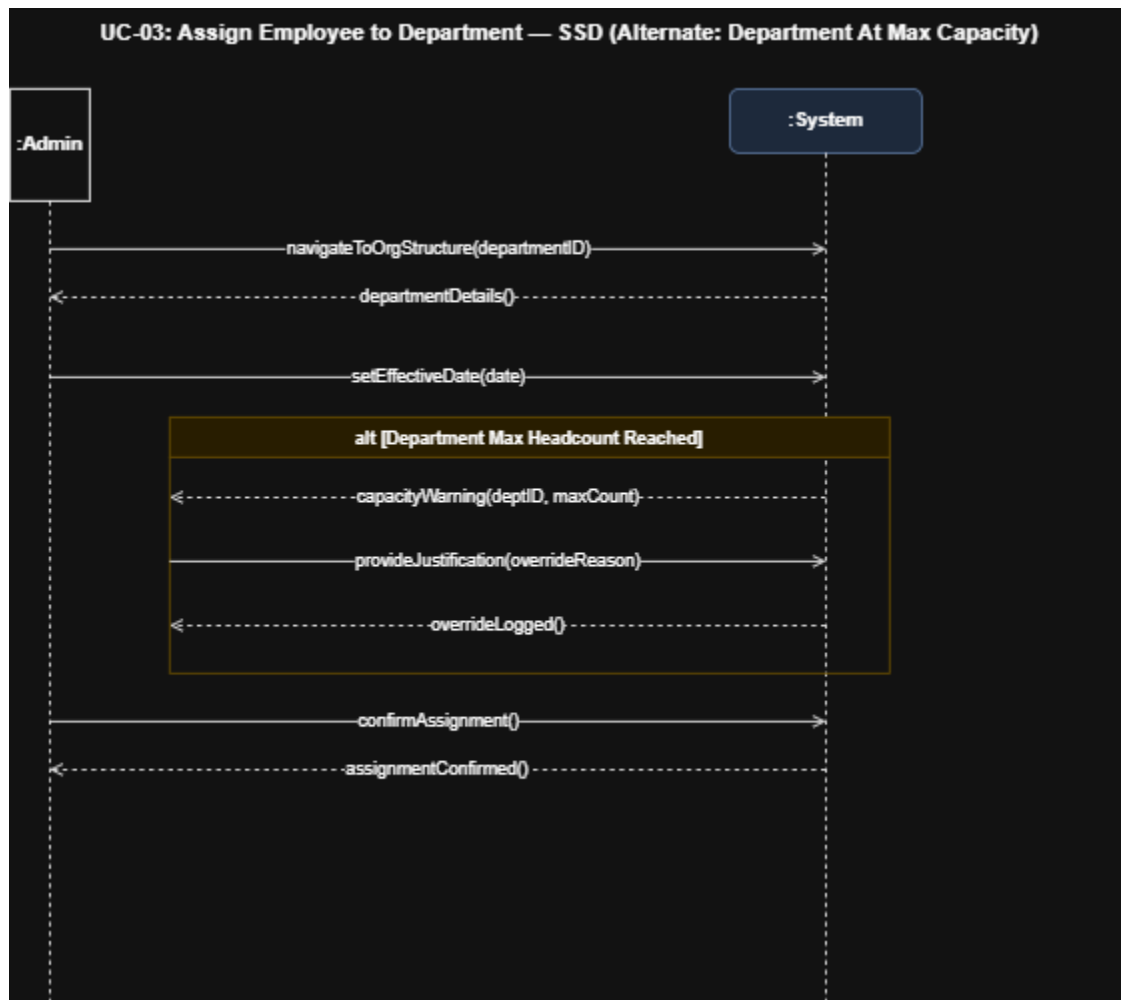
UC-03 — Assign Employee to Department

Use Case ID	UC-03
Use Case Name	Assign Employee to Department
Primary Actor	HR Officer / Admin
Description	Admin or HR Officer assigns or transfers an employee to a department with an effective date. System validates departmental capacity, updates the hierarchy, records history, and notifies managers.

Main Success Scenario SSD:



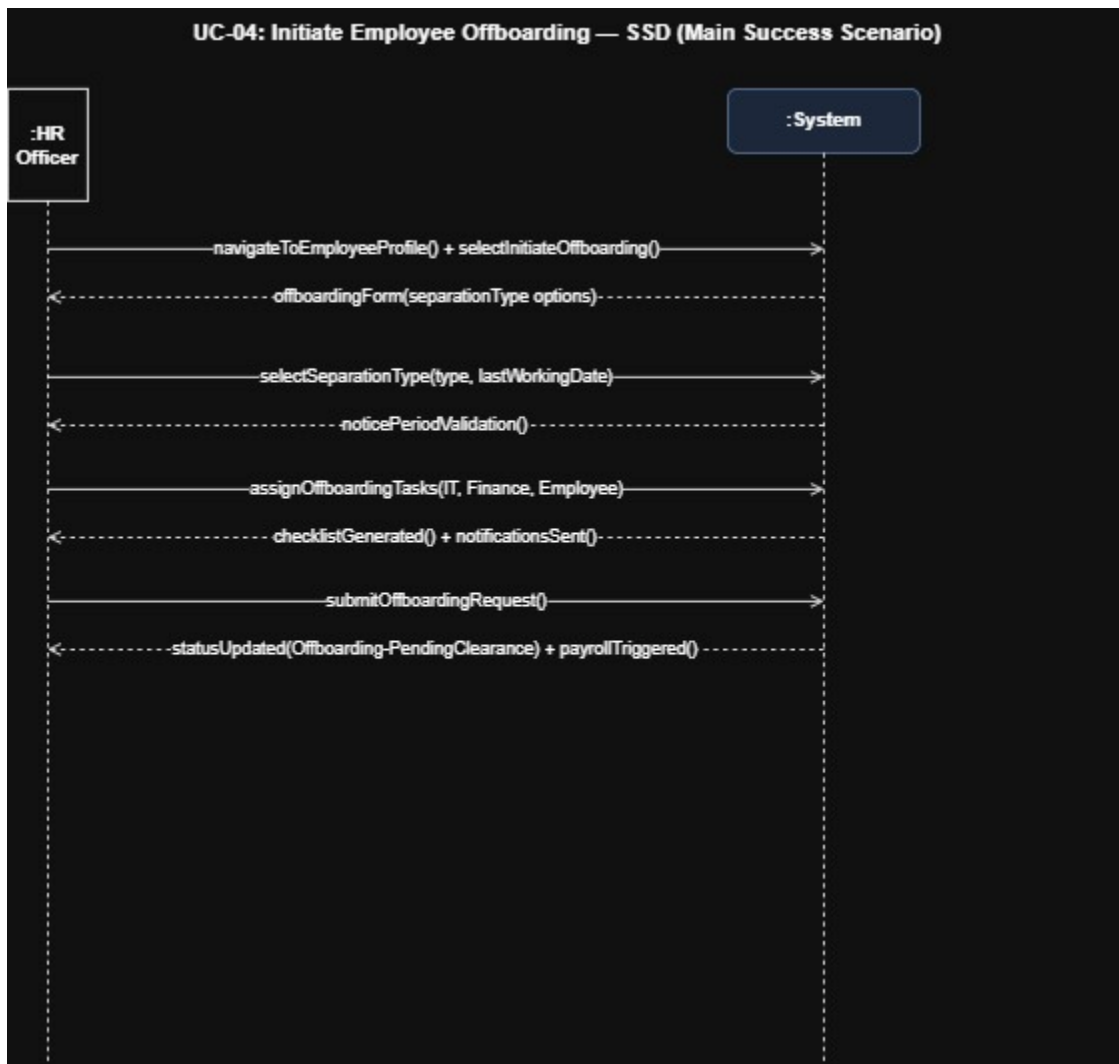
Major Alternative / Exception Flow SSDs:



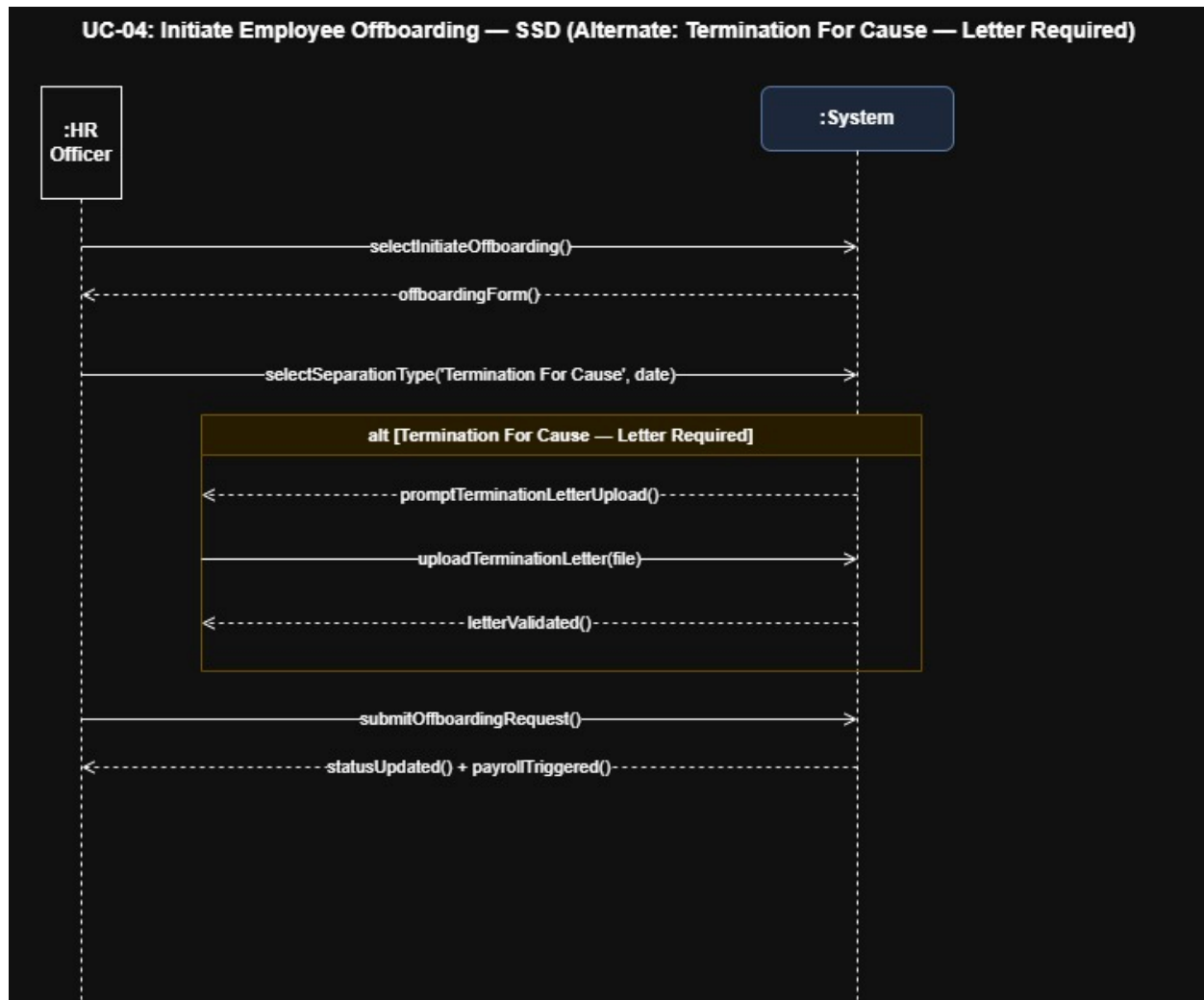
UC-04 — Initiate Employee Offboarding

Use Case ID	UC-04
Use Case Name	Initiate Employee Offboarding
Primary Actor	HR Officer
Description	HR Officer starts the offboarding workflow by specifying separation type, effective date, and exit reason. System creates a checklist, schedules access revocation, triggers final settlement, and updates employee status.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:



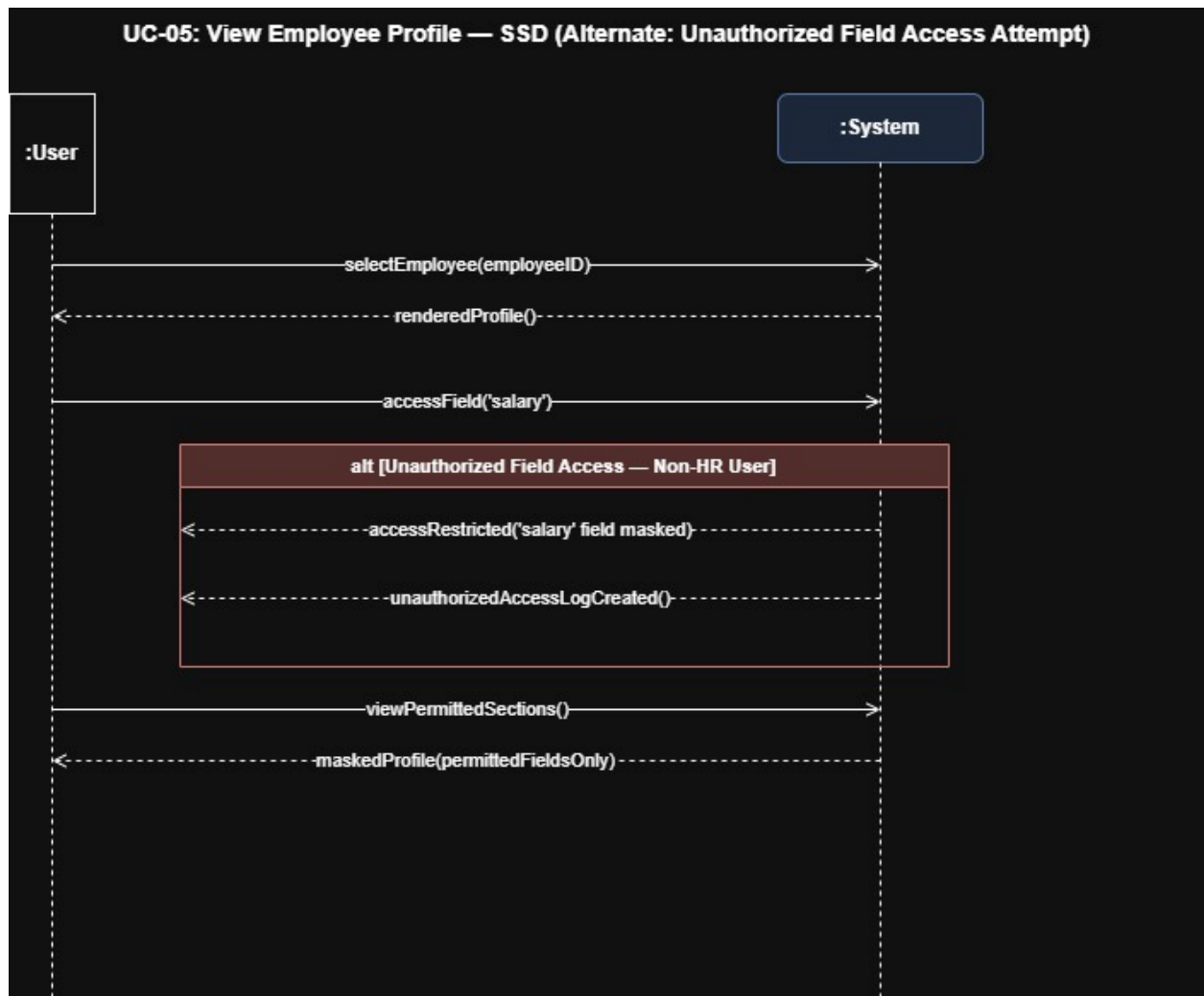
UC-05 — View Employee Profile

Use Case ID	UC-05
Use Case Name	View Employee Profile
Primary Actor	Employee / HR
Description	An authenticated user accesses a consolidated employee profile with personal details, employment history, leave balances, attendance summary, and performance ratings. Role-based field masking is enforced server-side.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:

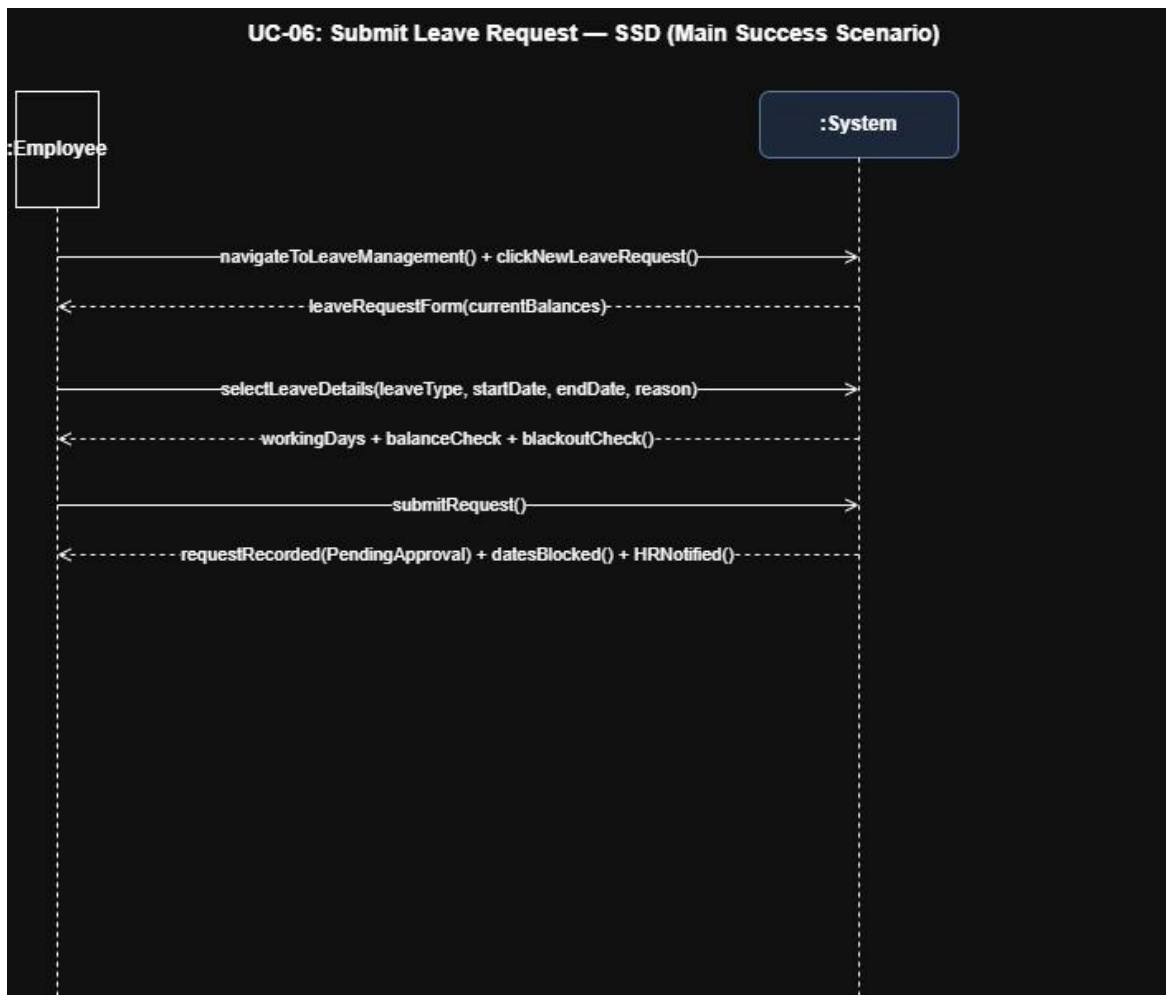


2.2 Leave and Attendance Management Module (UC-06 – UC-10)

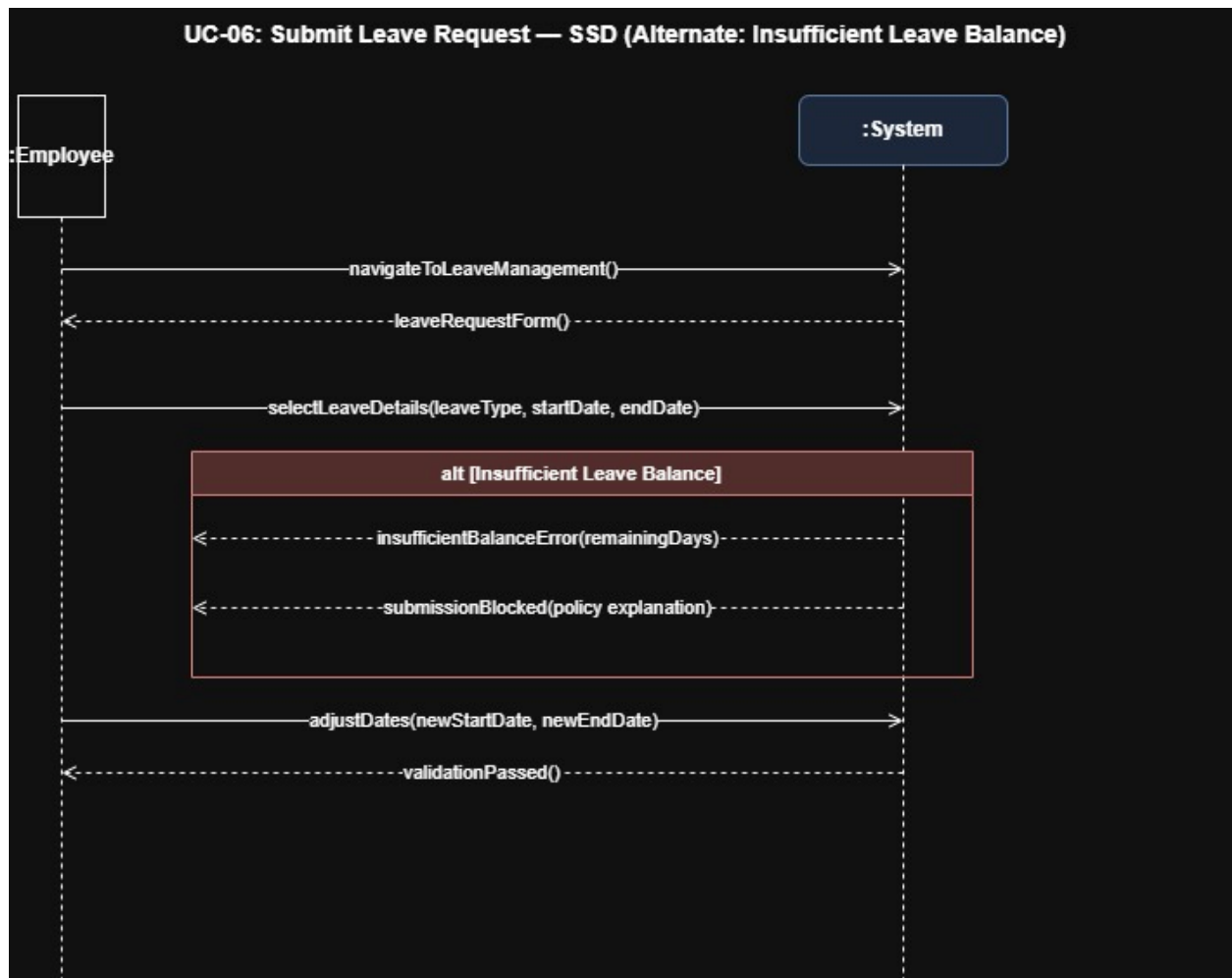
UC-06 — Submit Leave Request

Use Case ID	UC-06
Use Case Name	Submit Leave Request
Primary Actor	Employee
Description	Employee submits a formal leave request by selecting leave type and dates. System validates balance and policy calendar, records the request as Pending Approval, blocks dates in the attendance calendar, and notifies HR.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:



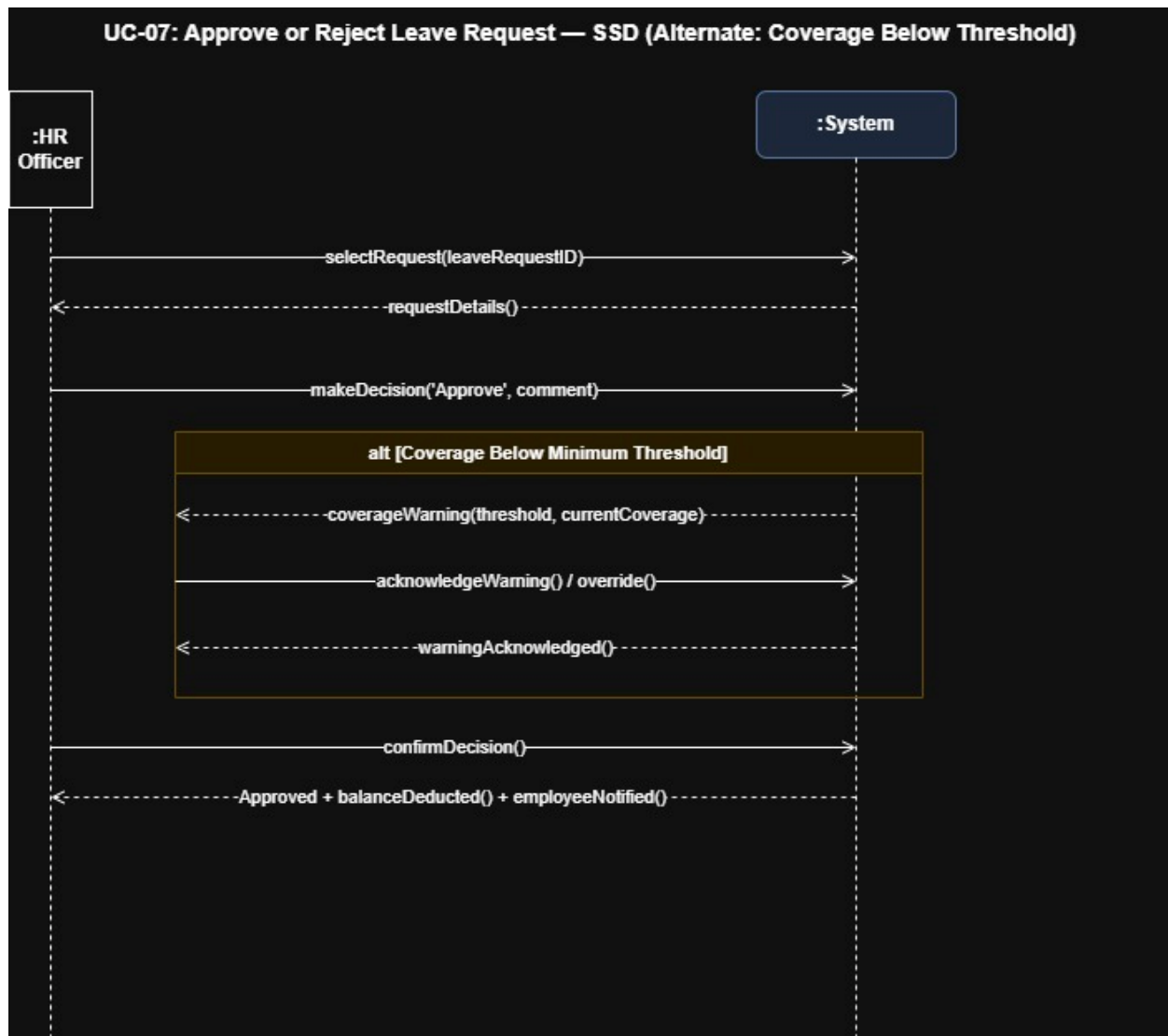
UC-07 — Approve or Reject Leave Request

Use Case ID	UC-07
Use Case Name	Approve or Reject Leave Request
Primary Actor	HR Officer
Description	HR Officer reviews a pending leave request and approves or rejects it with a mandatory comment. On approval, system deducts leave balance, updates attendance calendar, and notifies employee.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:



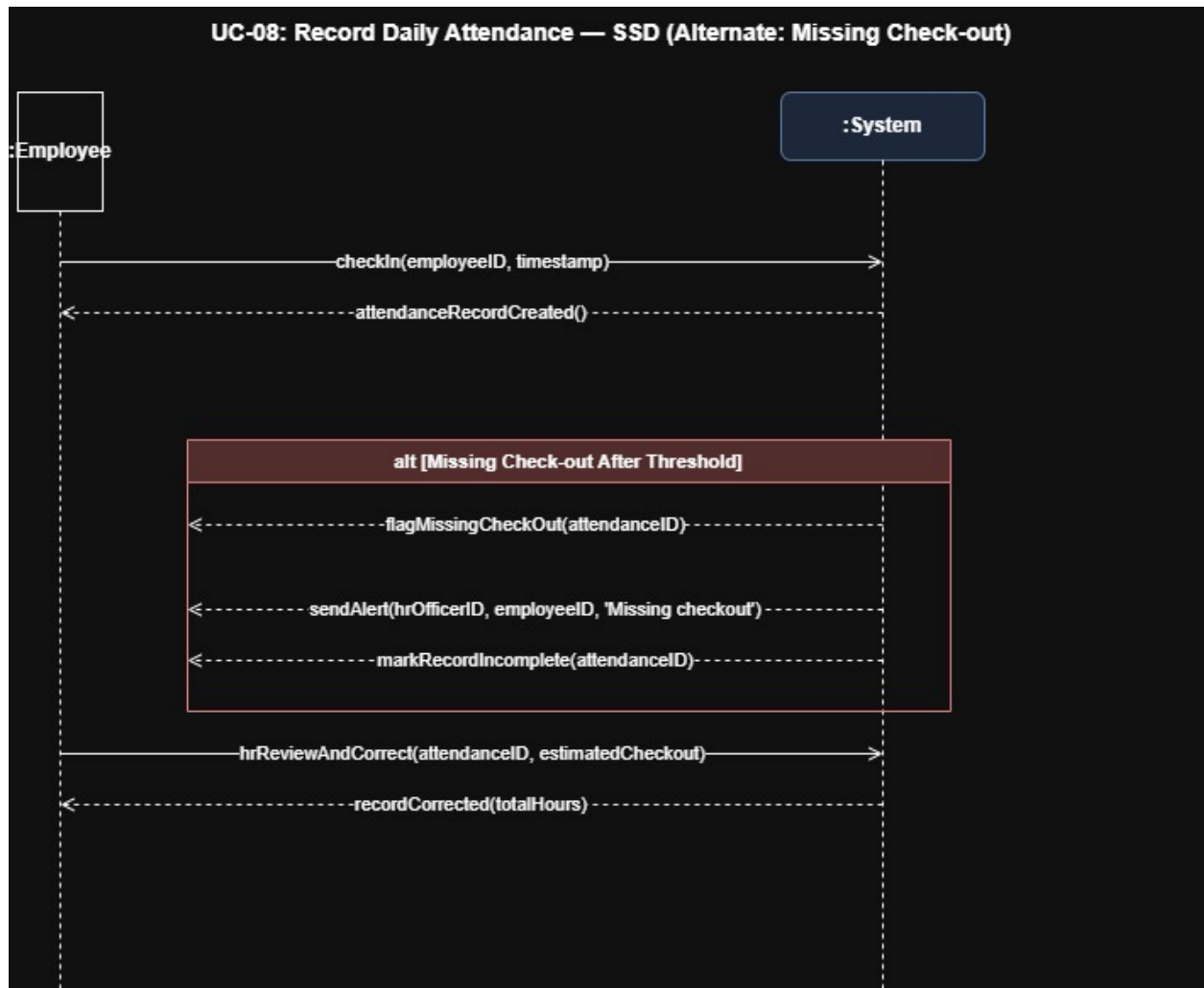
UC-08 — Record Daily Attendance

Use Case ID	UC-08
Use Case Name	Record Daily Attendance
Primary Actor	Employee
Description	System records check-in and check-out timestamps via biometric device, web portal, or mobile app. Computes work hours, flags late arrivals and early departures, and stores attendance linked to payroll.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:



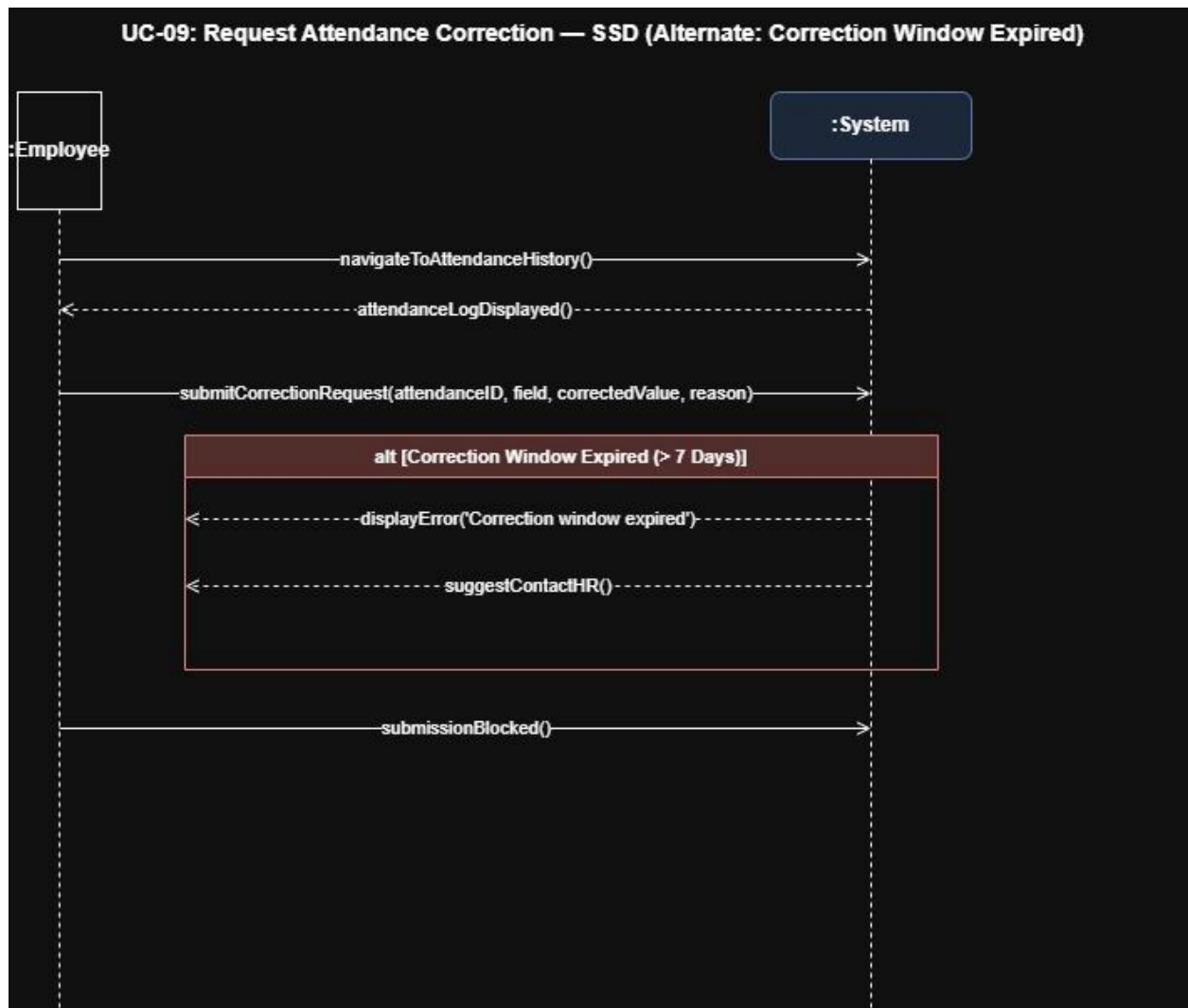
UC-09 — Request Attendance Correction

Use Case ID	UC-09
Use Case Name	Request Attendance Correction
Primary Actor	Employee
Description	Employee submits a correction request specifying affected date, incorrect value, corrected value, and reason. HR reviews and approved corrections update the record with an audit annotation.

Main Success Scenario SSD:



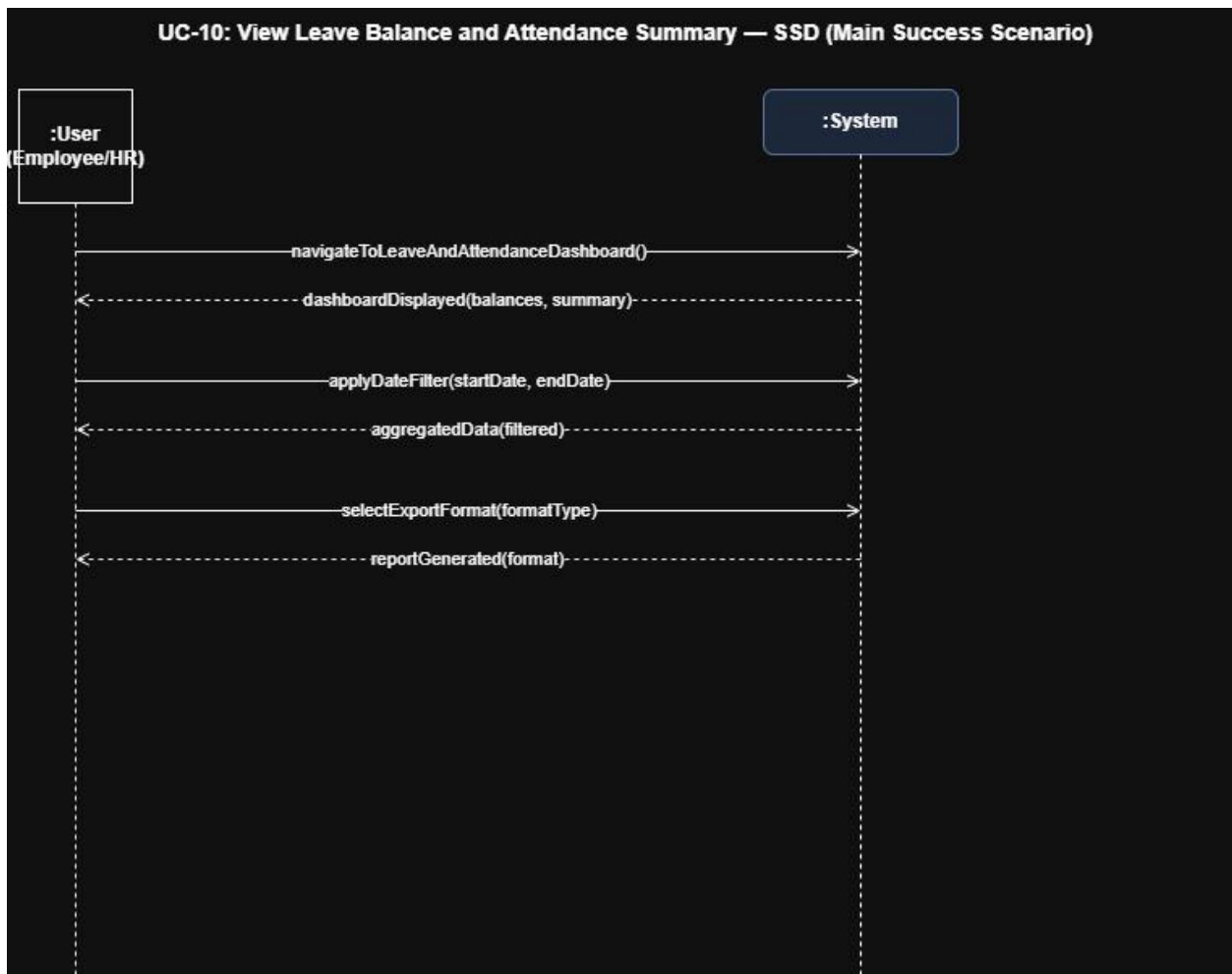
Major Alternative / Exception Flow SSDs:



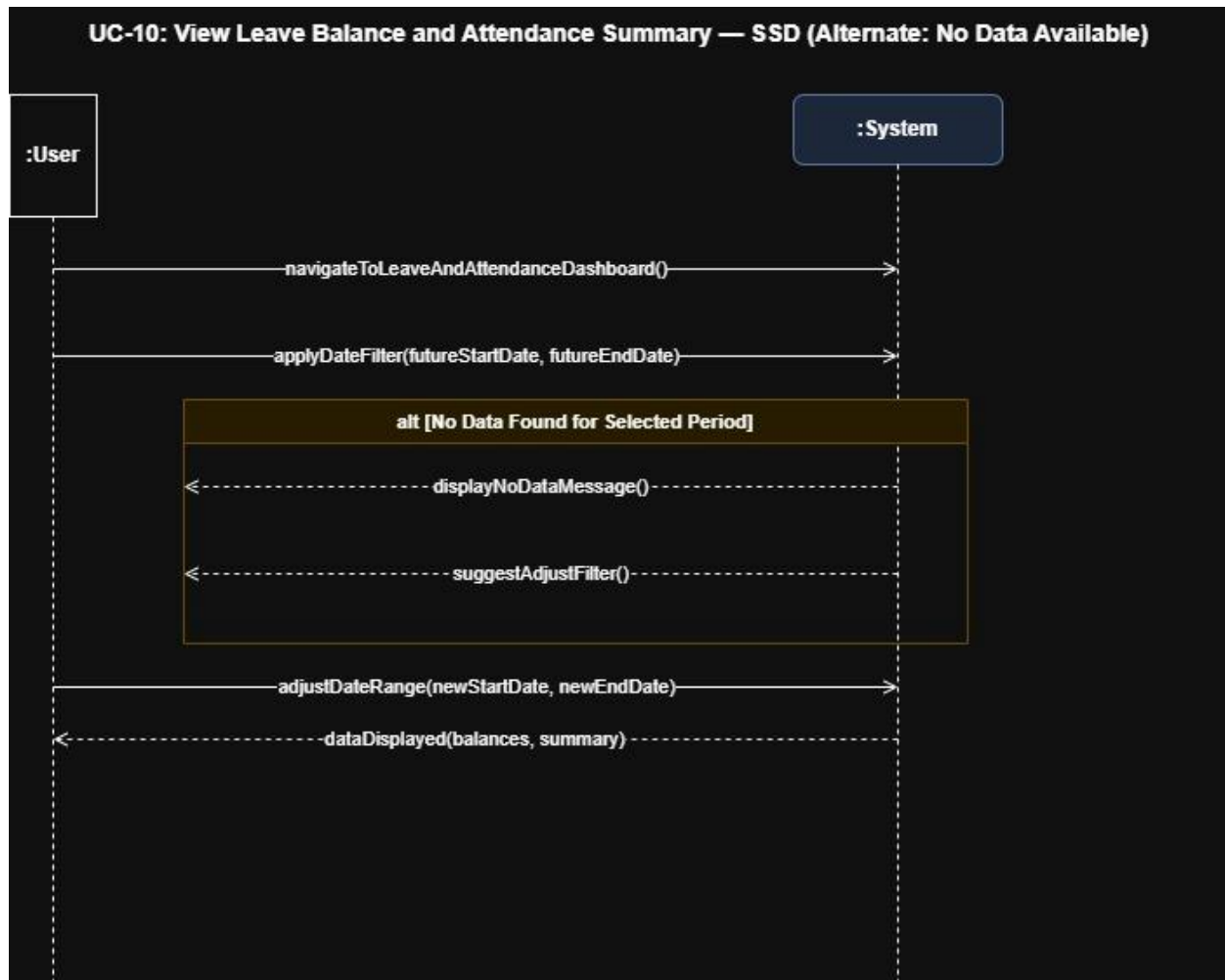
UC-10 — View Leave Balance and Attendance Summary

Use Case ID	UC-10
Use Case Name	View Leave Balance and Attendance Summary
Primary Actor	Employee / HR
Description	Authorized users access a real-time dashboard showing remaining leave balances by type and monthly attendance KPIs. System aggregates data from leave and attendance modules and supports export to PDF or Excel.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:

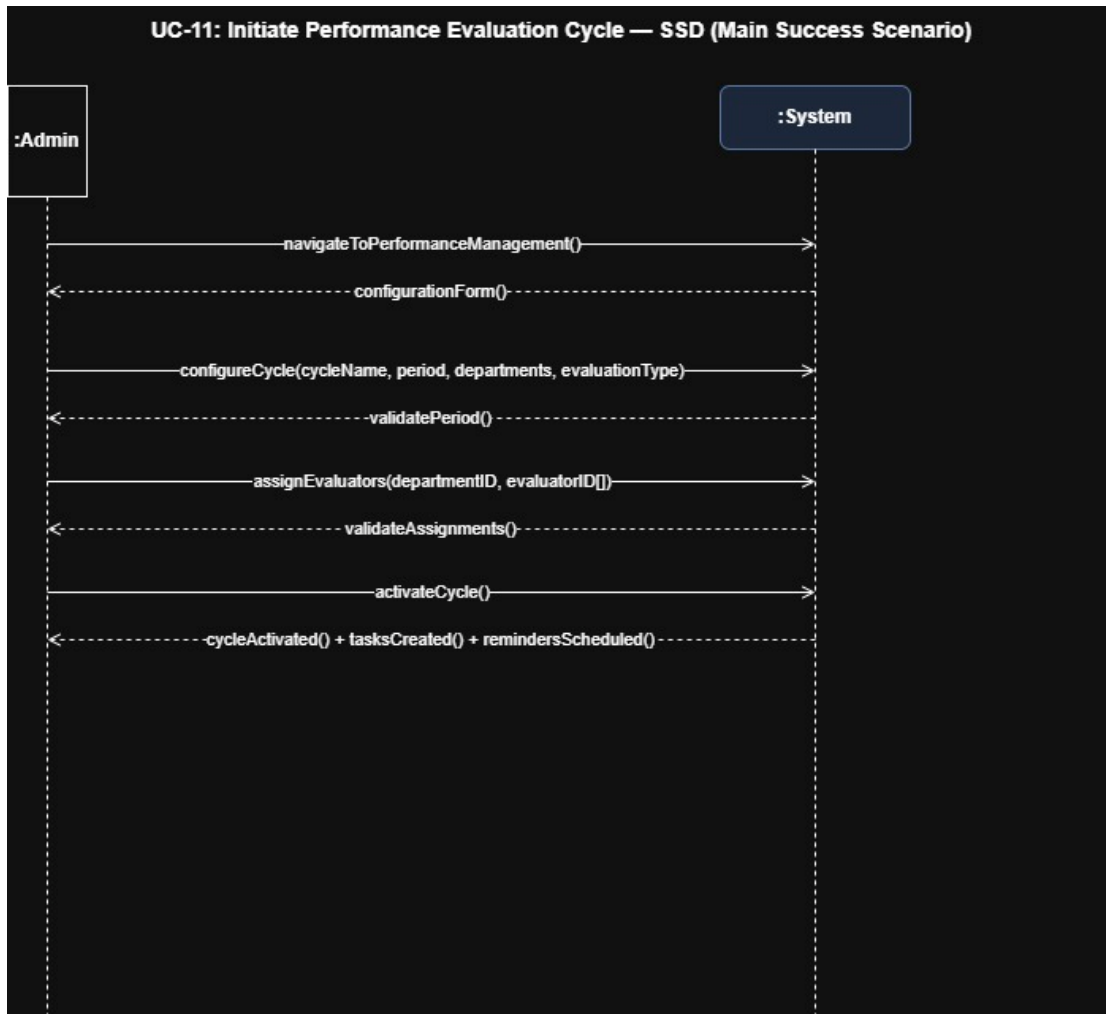


2.3 Performance and Compliance Management Module (UC-11 – UC-15)

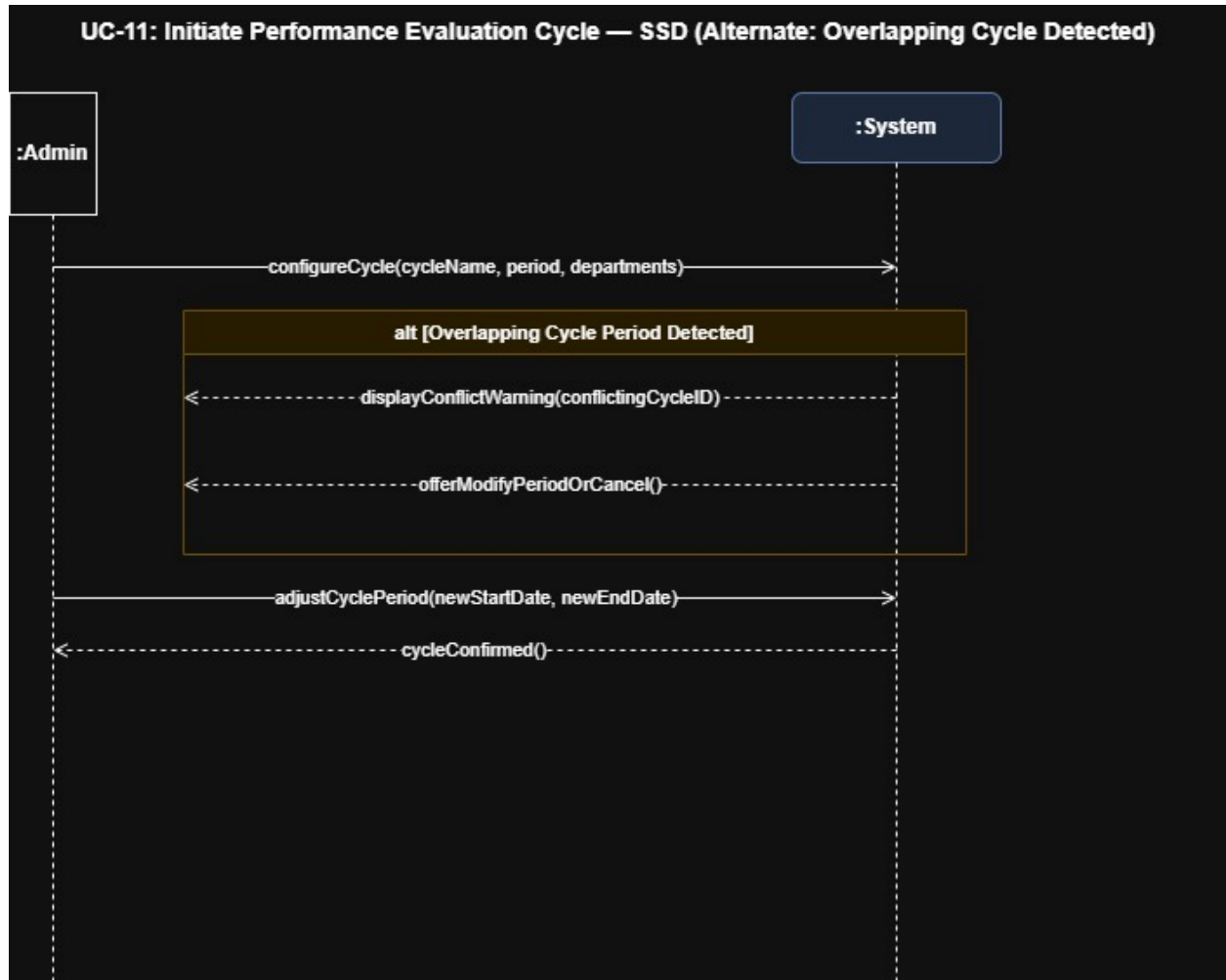
UC-11 — Initiate Performance Evaluation Cycle

Use Case ID	UC-11
Use Case Name	Initiate Performance Evaluation Cycle
Primary Actor	Admin
Description	Admin configures and launches a performance evaluation cycle by defining period, scope, evaluators, and deadlines. System distributes tasks, opens the evaluation module, and schedules automated reminder notifications.

Main Success Scenario SSD:



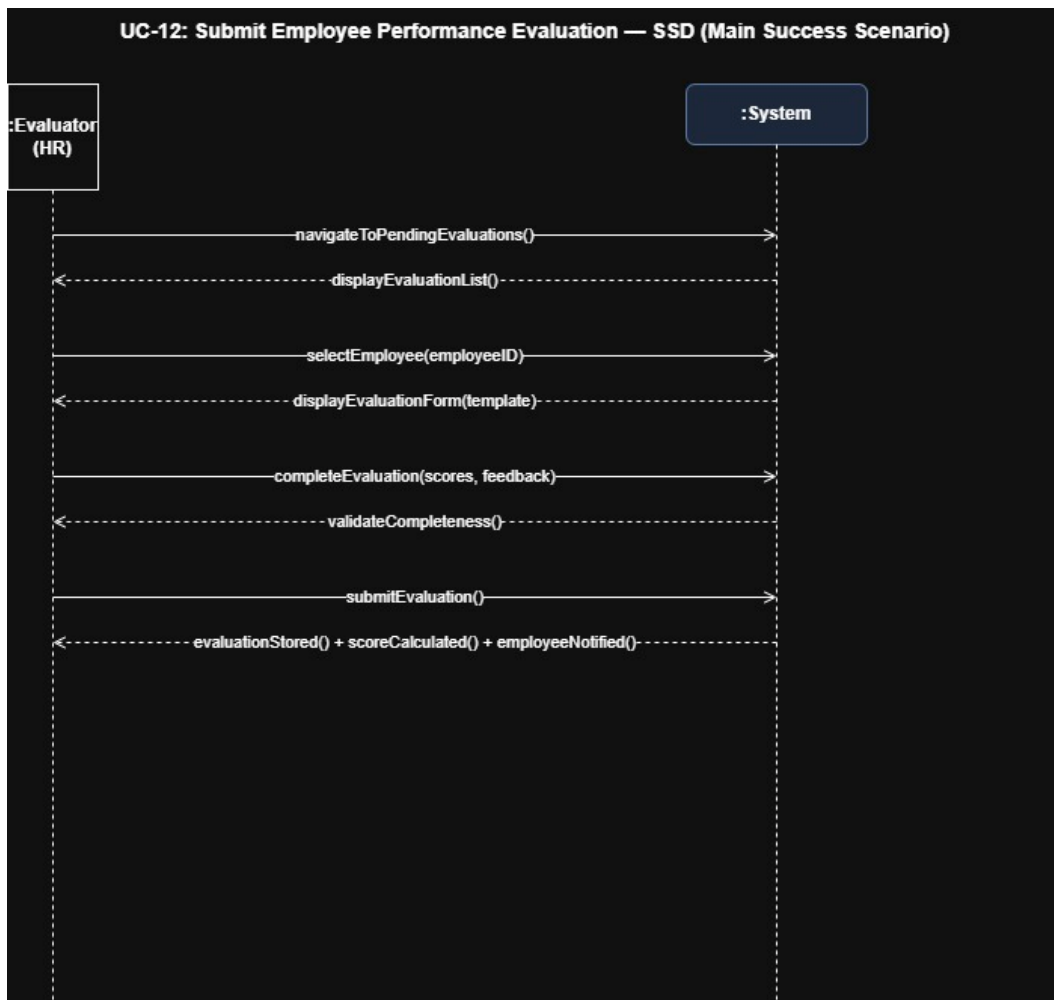
Major Alternative / Exception Flow SSDs:



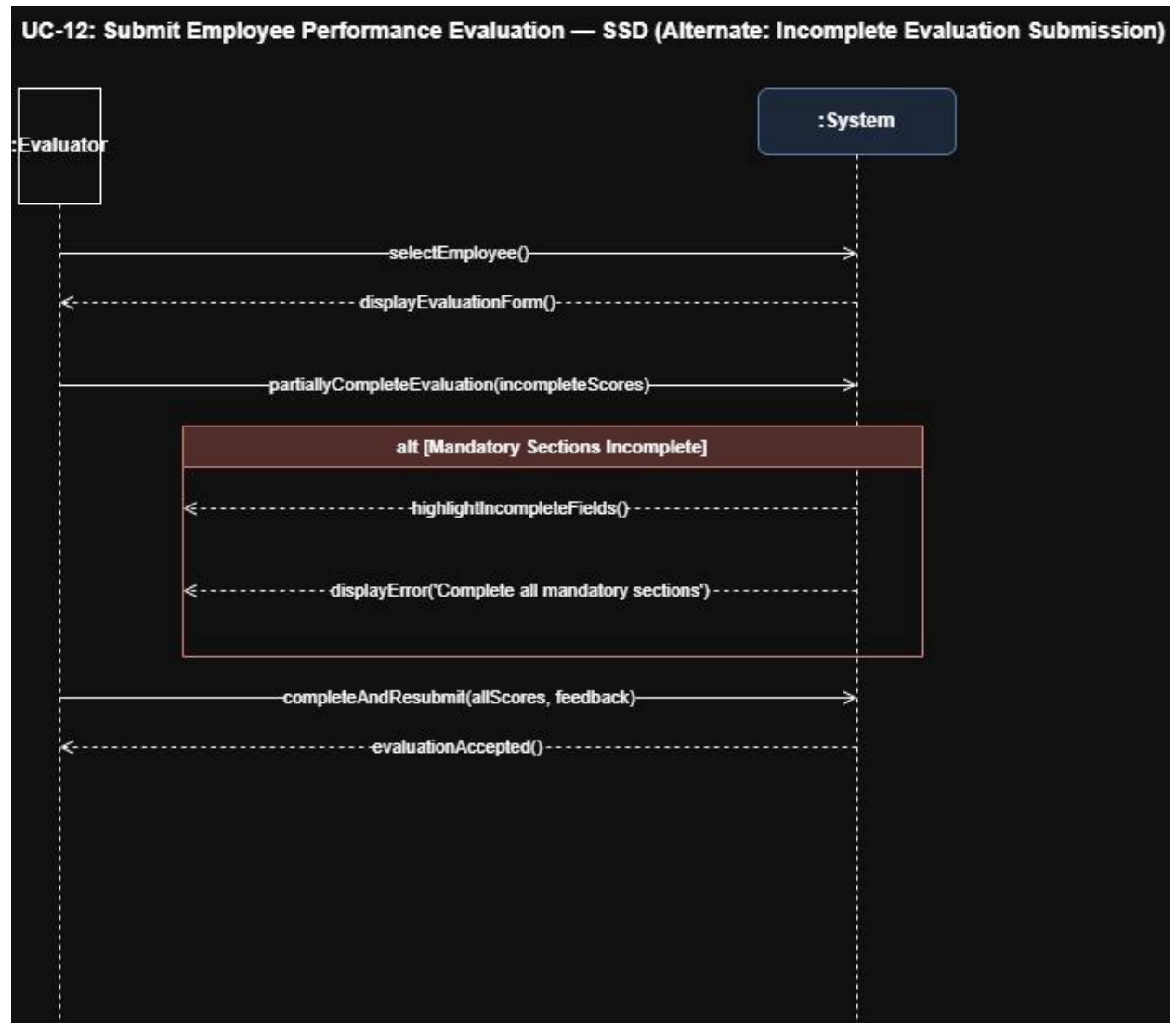
UC-12 — Submit Employee Performance Evaluation

Use Case ID	UC-12
Use Case Name	Submit Employee Performance Evaluation
Primary Actor	HR / Evaluator
Description	Assigned evaluator completes a structured performance evaluation form for an employee during an active evaluation cycle by scoring competencies and KPIs. System validates completeness, calculates aggregate score, and records in history.

Main Success Scenario SSD:



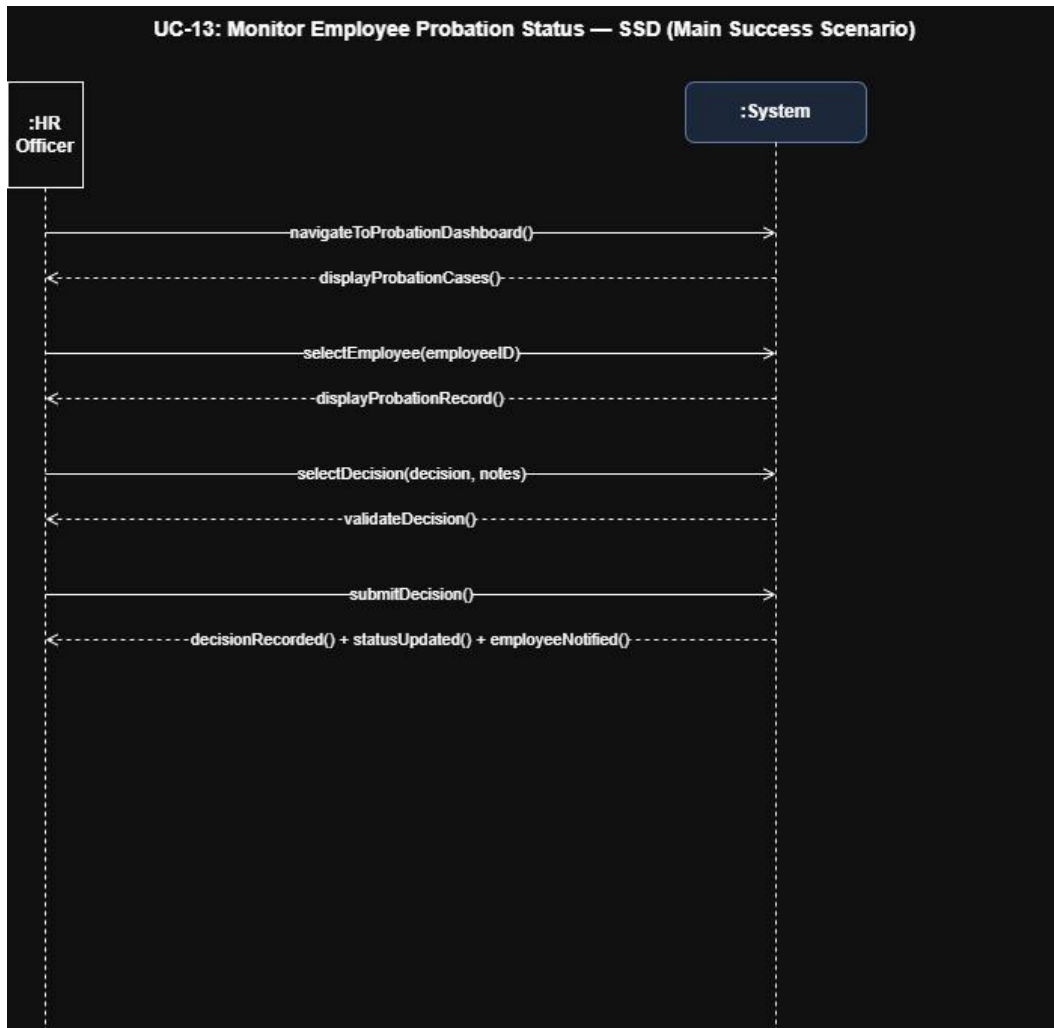
Major Alternative / Exception Flow SSDs:



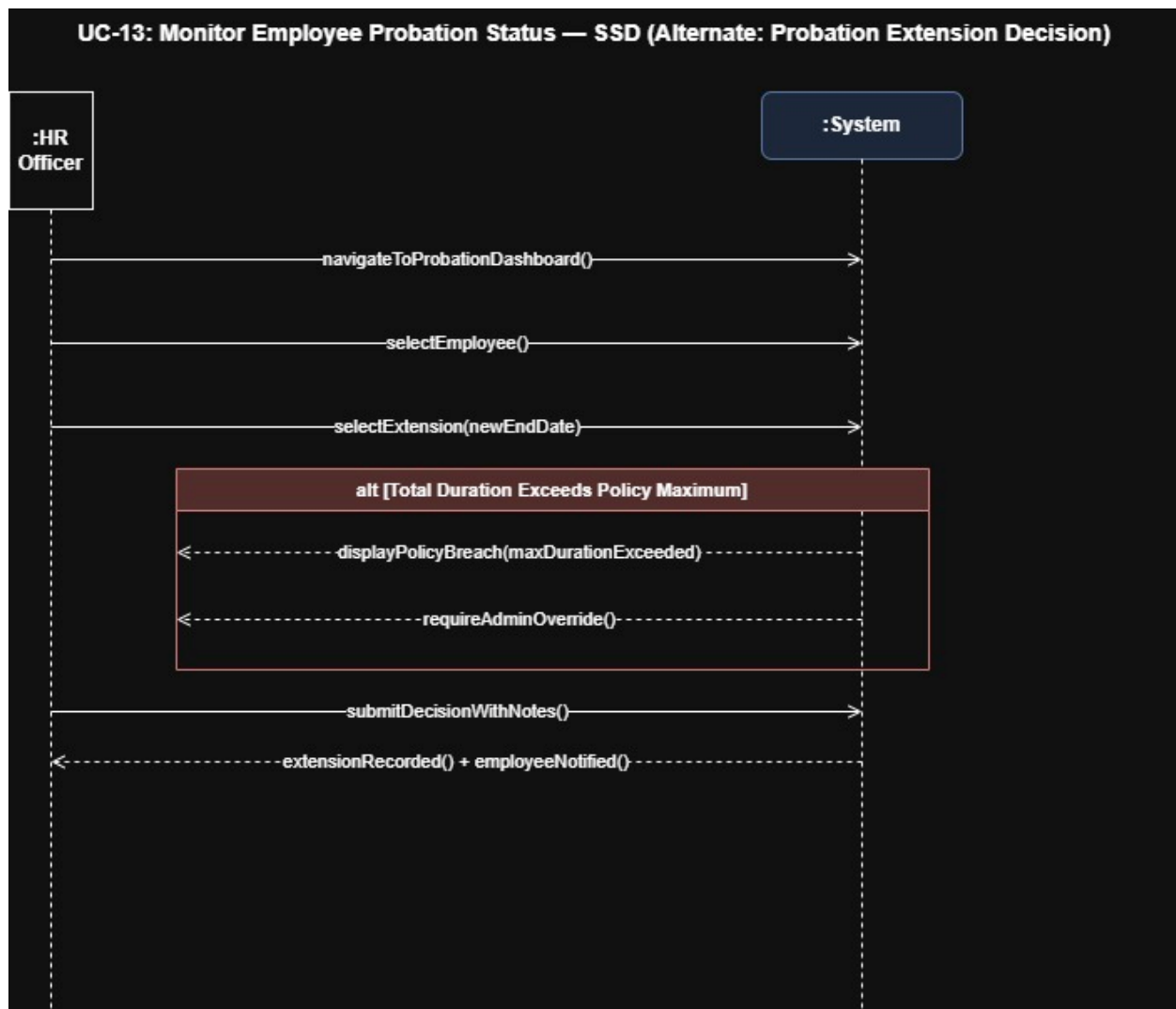
UC-13 — Monitor Employee Probation Status

Use Case ID	UC-13
Use Case Name	Monitor Employee Probation Status
Primary Actor	HR Officer
Description	HR Officer monitors employees on probation, tracking evaluation milestones and decision dates. System sends automated deadline alerts and allows HR to record confirmation, extension, or termination decisions.

Main Success Scenario SSD:



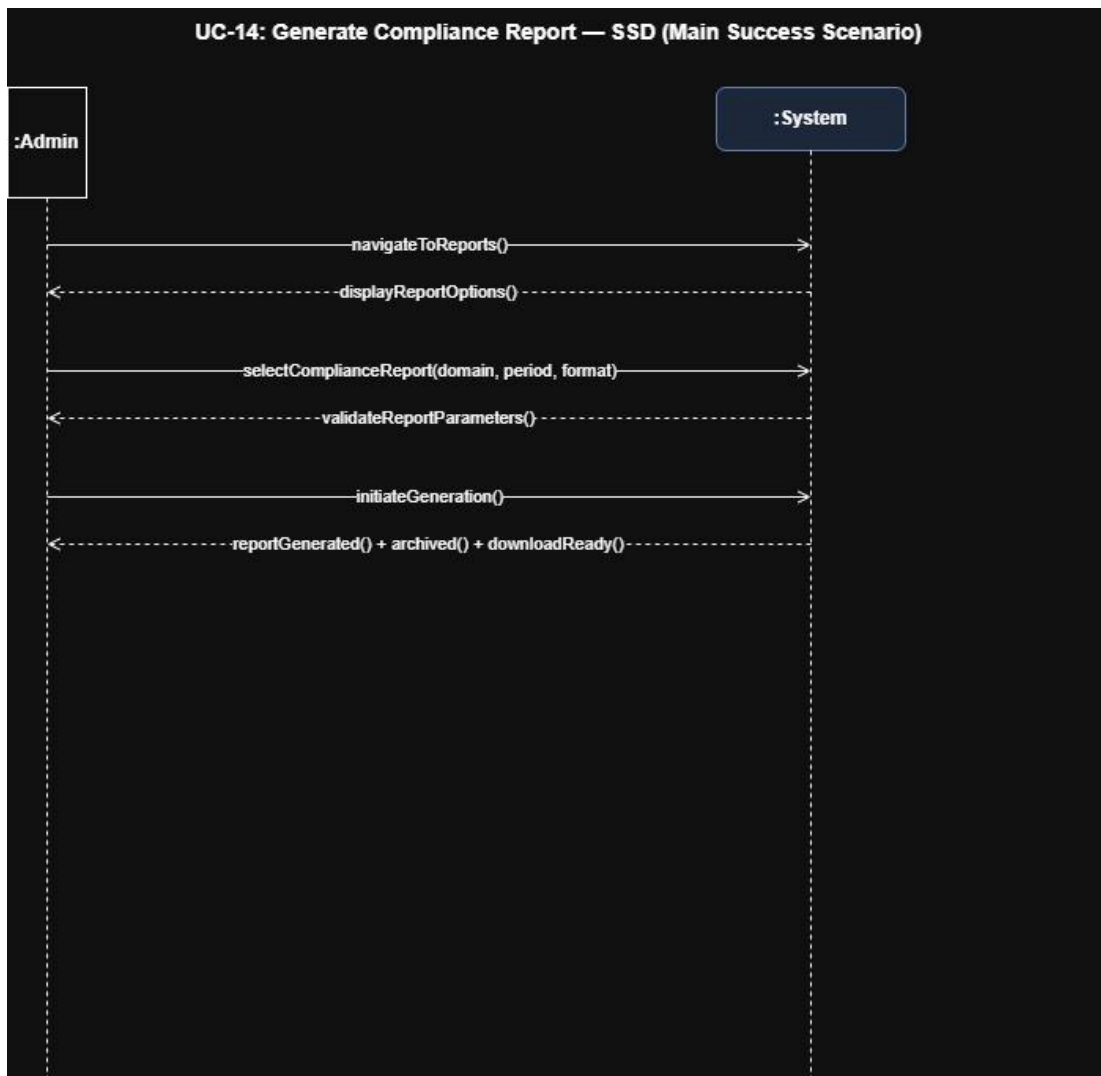
Major Alternative / Exception Flow SSDs:



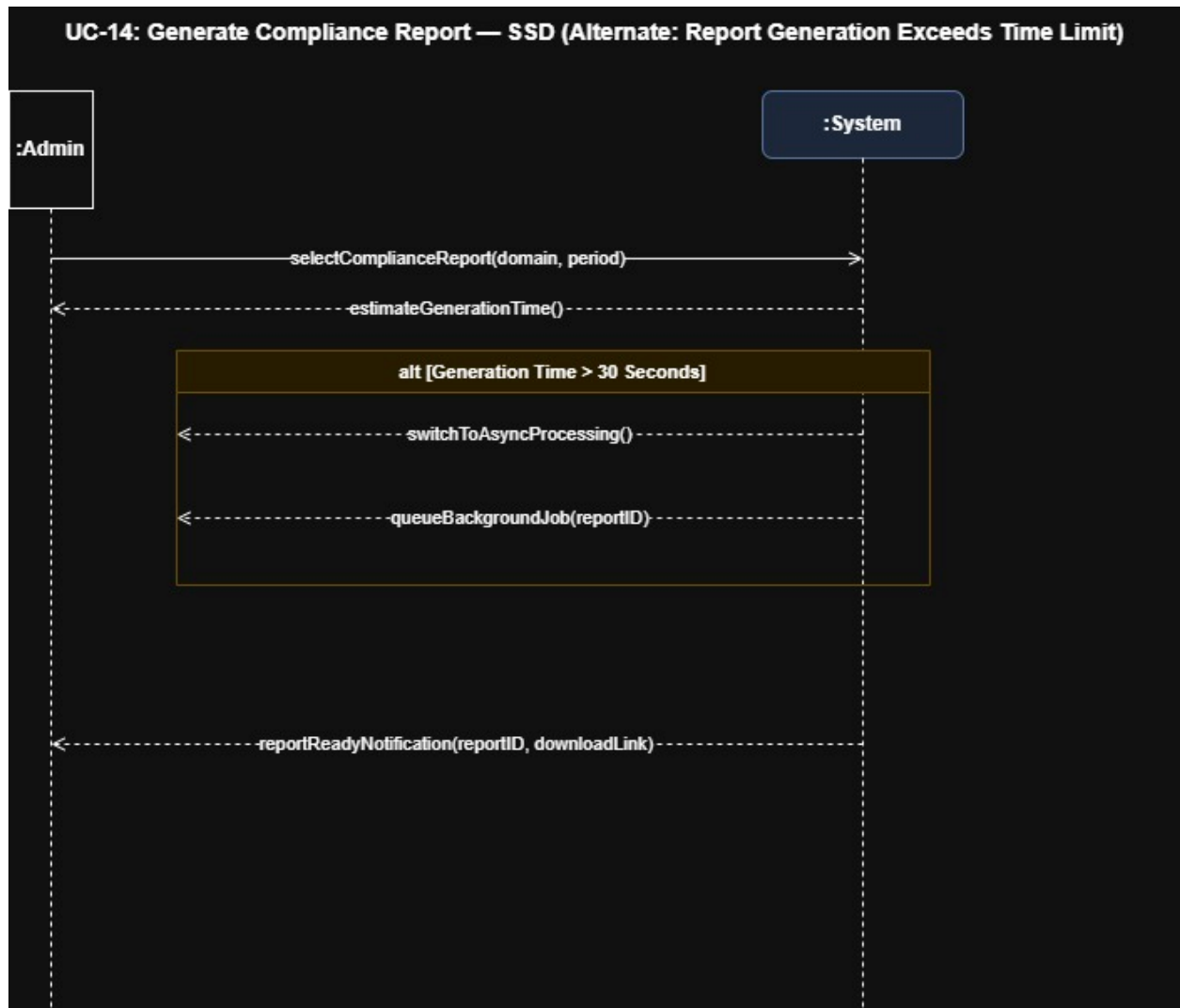
UC-14 — Generate Compliance Report

Use Case ID	UC-14
Use Case Name	Generate Compliance Report
Primary Actor	Admin
Description	Admin generates a formal compliance report for a specified period and compliance domain. System aggregates policy adherence data, applies regulatory filters, formats time-stamped output, and archives it for a minimum of five years.

Main Success Scenario SSD:



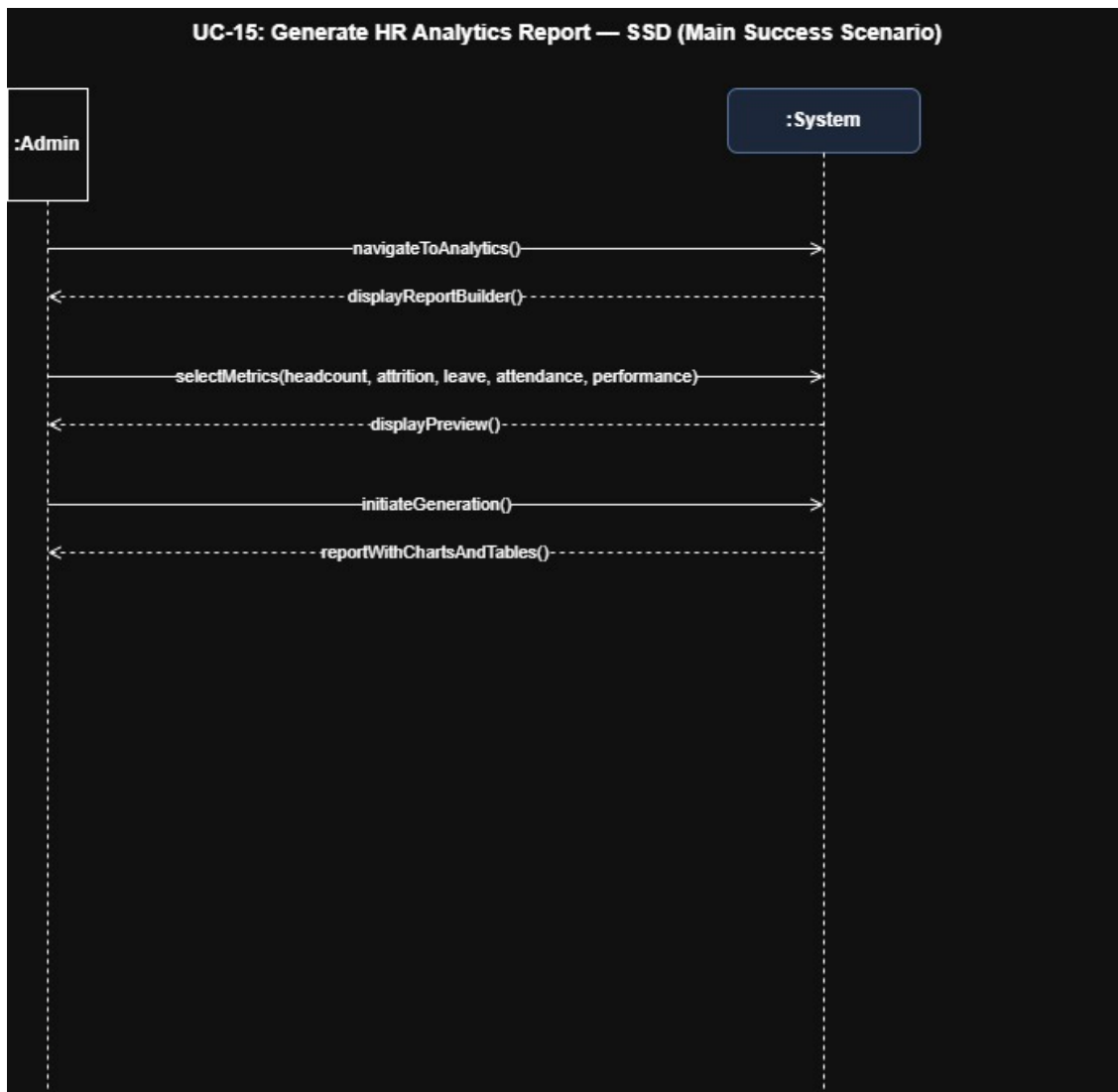
Major Alternative / Exception Flow SSDs:



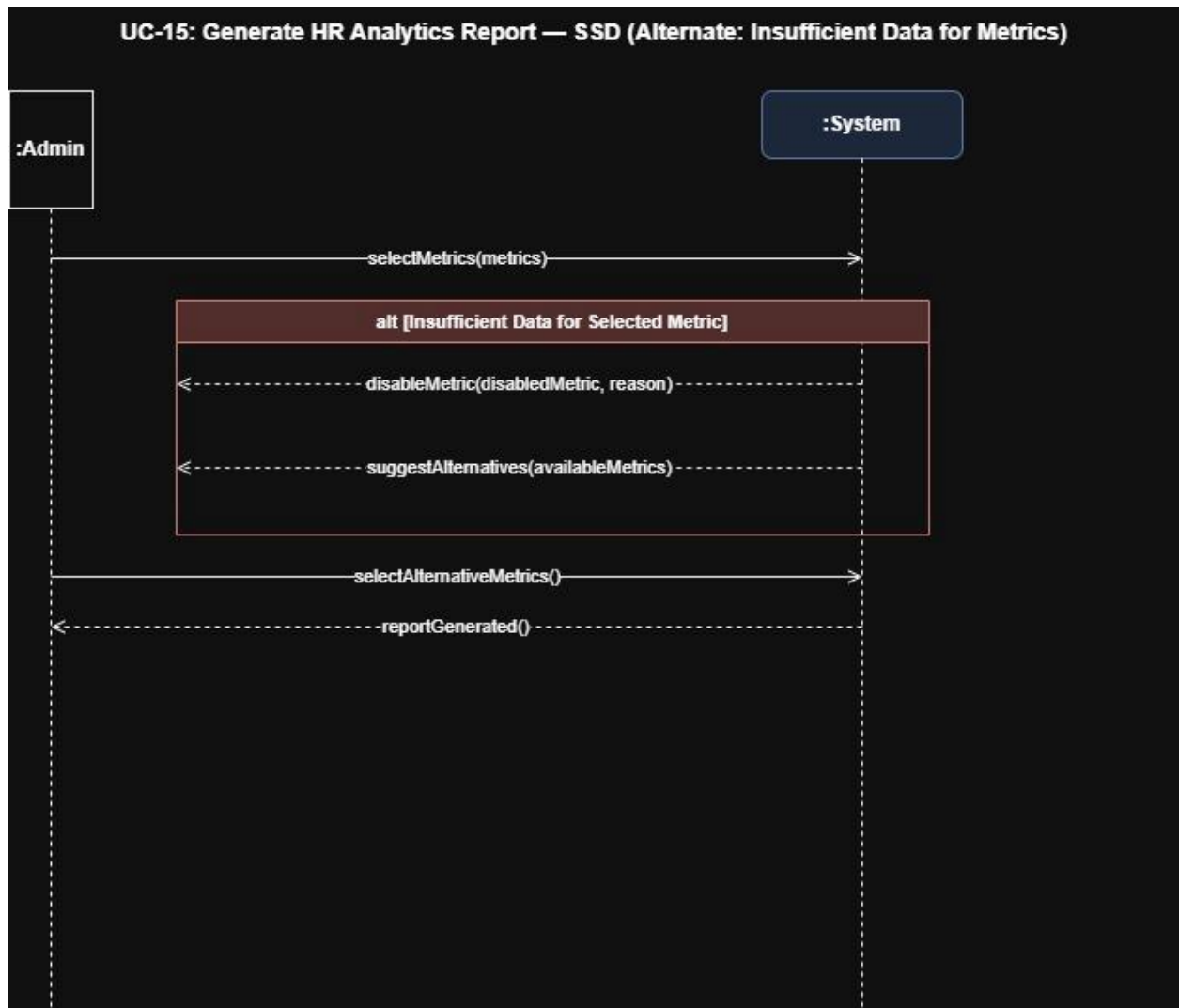
UC-15 — Generate HR Analytics Report

Use Case ID	UC-15
Use Case Name	Generate HR Analytics Report
Primary Actor	Admin
Description	Admin generates an analytics report covering workforce KPIs such as headcount, attrition, leave utilization, and performance benchmarks. System queries all HR modules, applies statistical aggregation, and renders interactive charts and data tables.

Main Success Scenario SSD:



Major Alternative / Exception Flow SSDs:

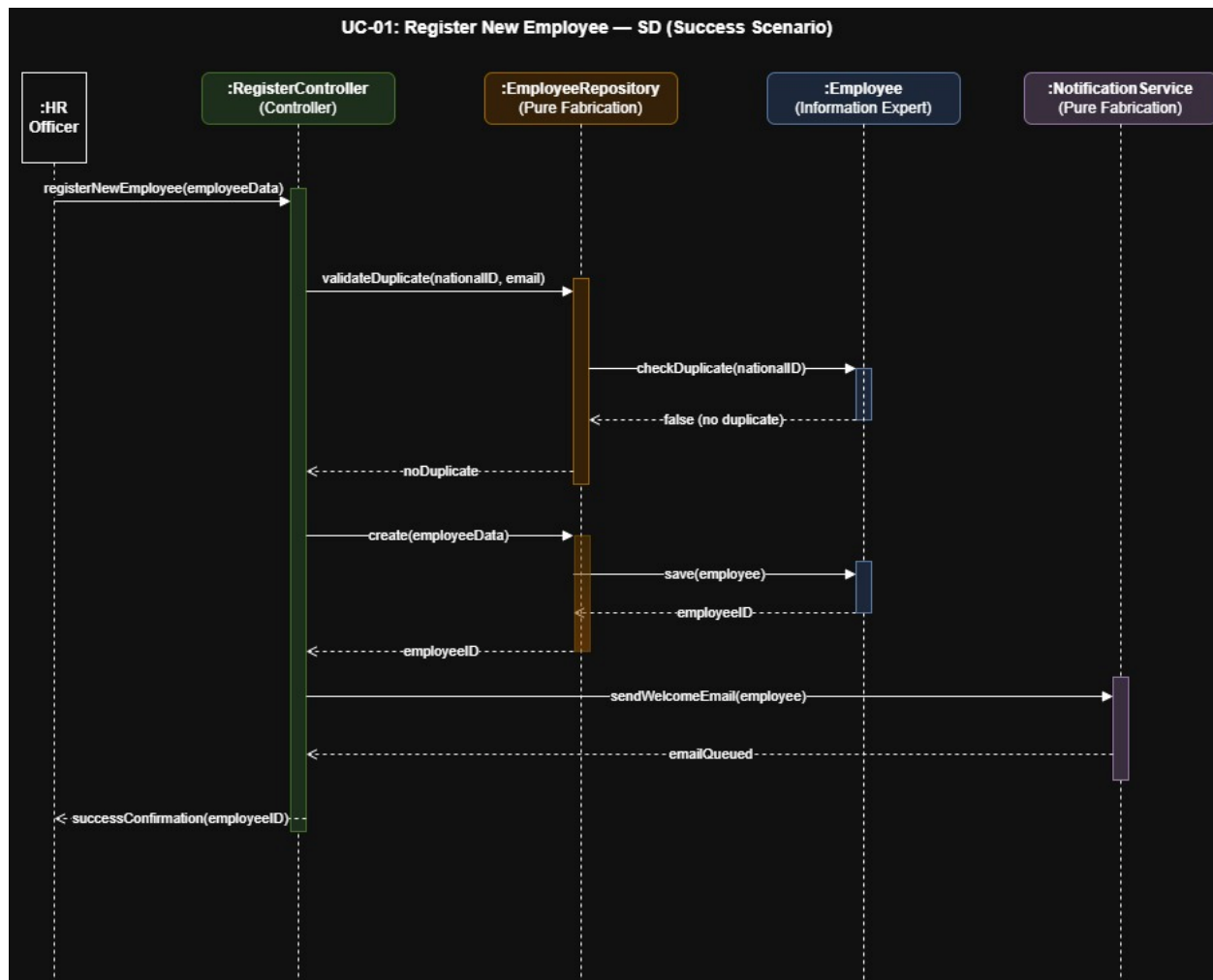


3. Sequence Diagrams (SDs)

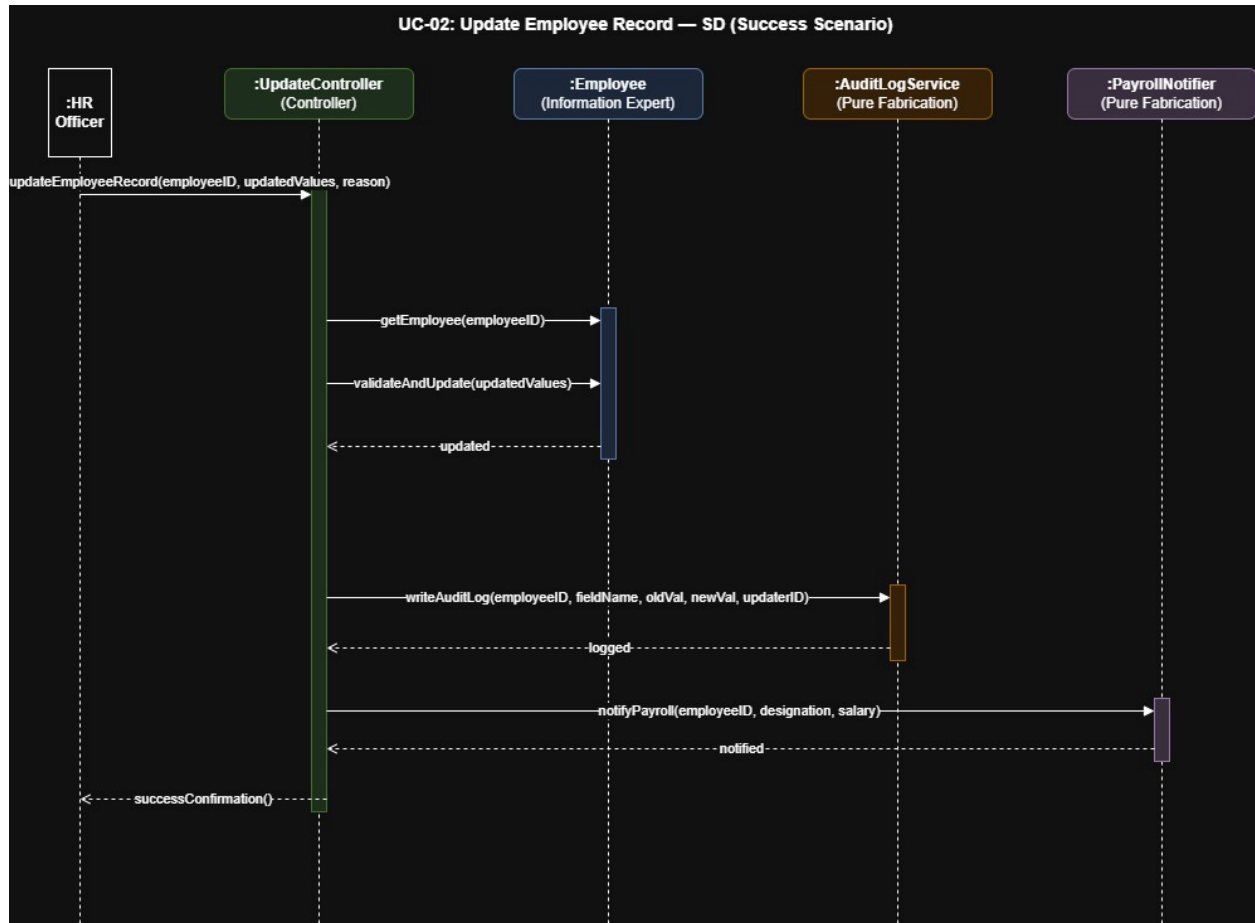
This section presents individual Sequence Diagrams (SDs) for each system event or operation identified in the SSDs. Each operation is annotated with a GRASP pattern note explaining the rationale for the responsibility assignment. Where applicable, Gang of Four (GoF) design patterns are also indicated.

3.1 Employee Management Module (UC-01 – UC-05)

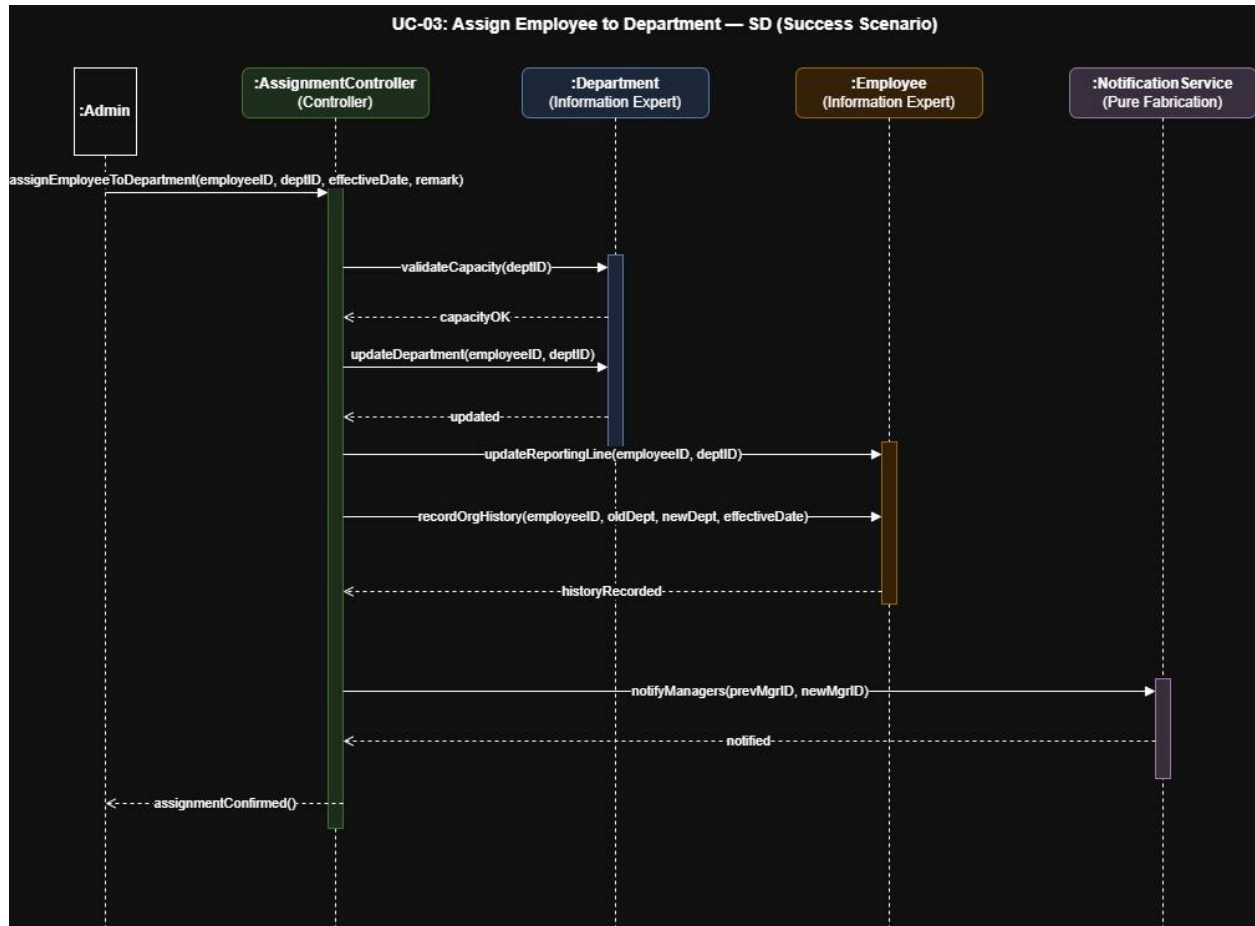
UC-01 – Sequence Diagrams



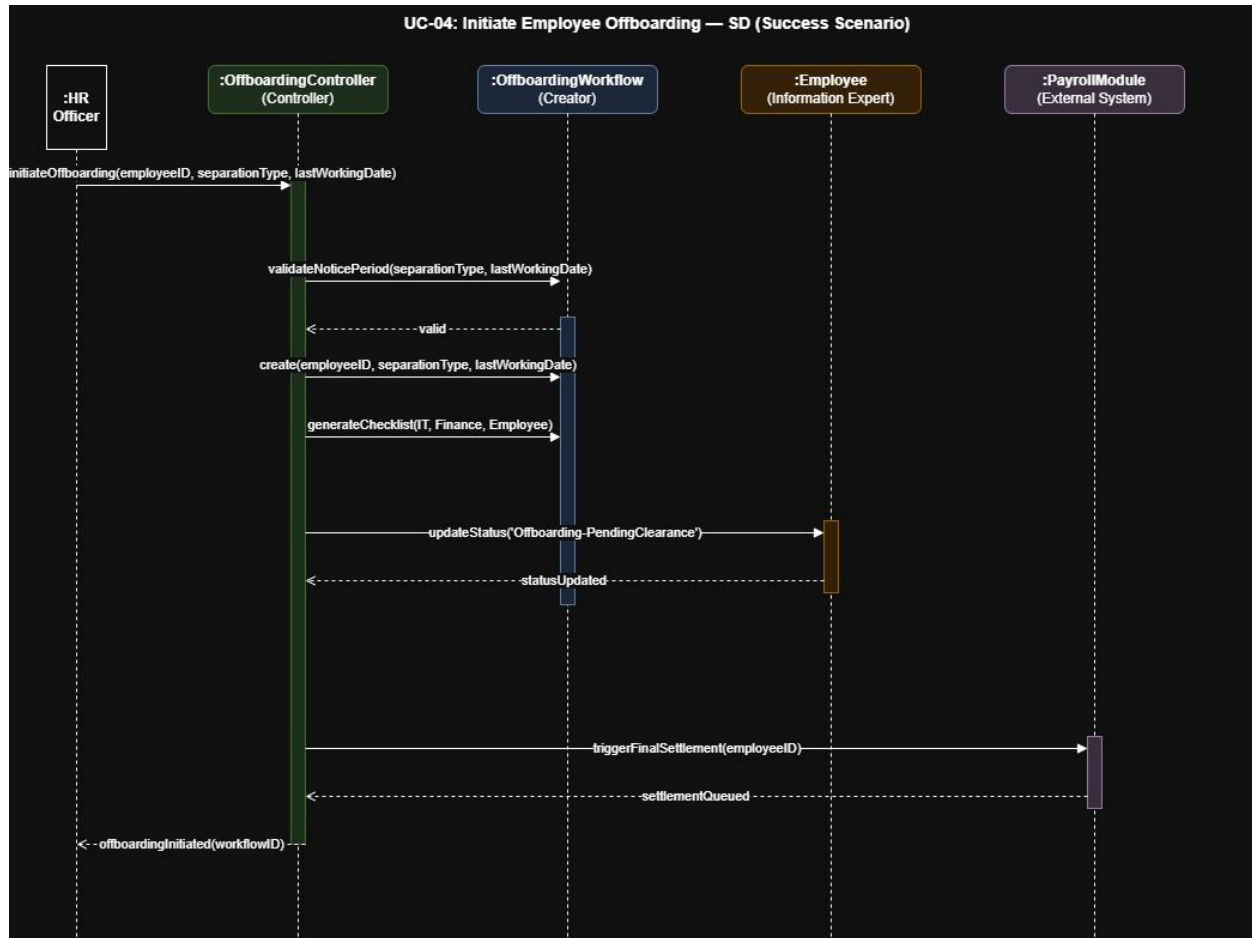
UC-02 – Sequence Diagrams



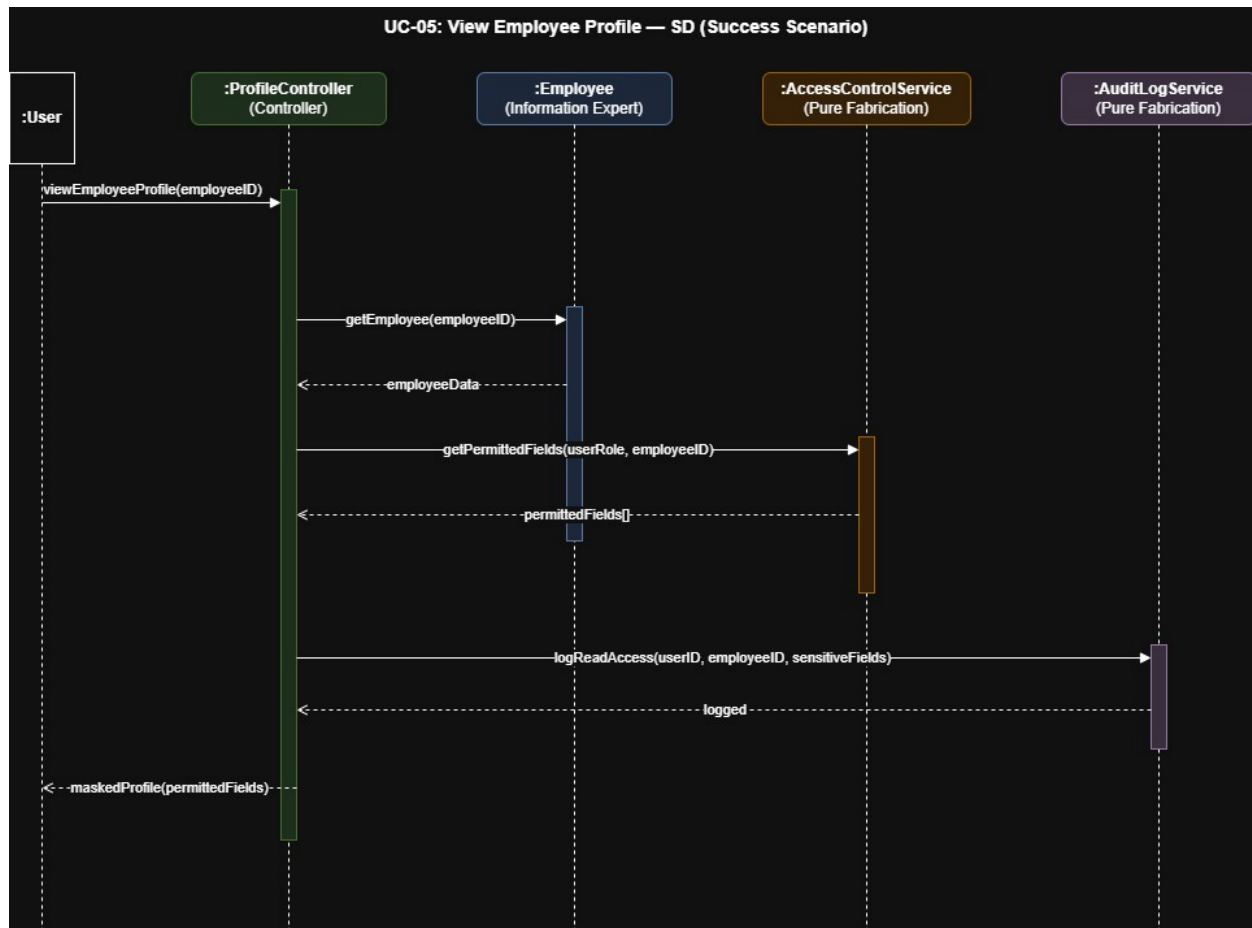
UC-03 – Sequence Diagrams



UC-04 – Sequence Diagrams

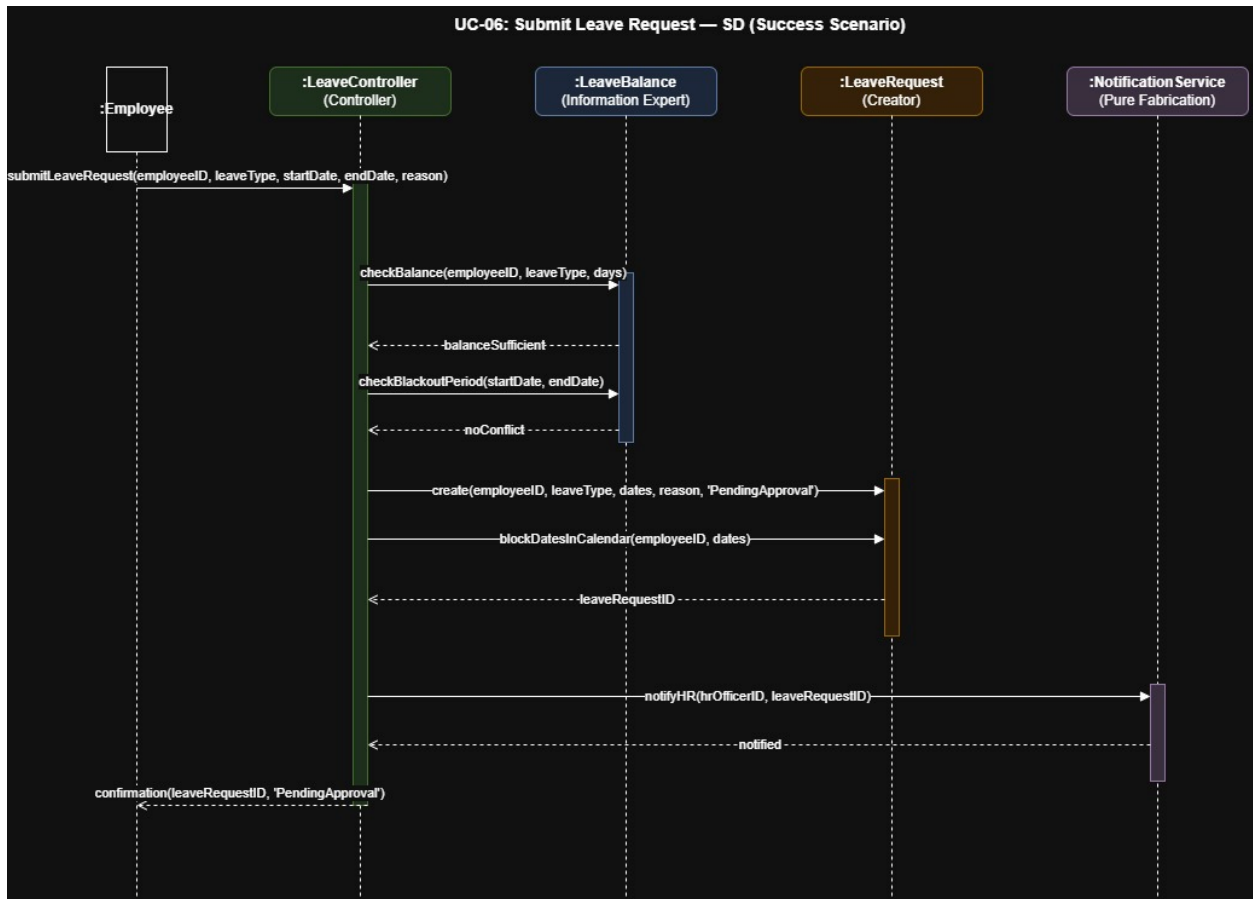


UC-05 – Sequence Diagrams

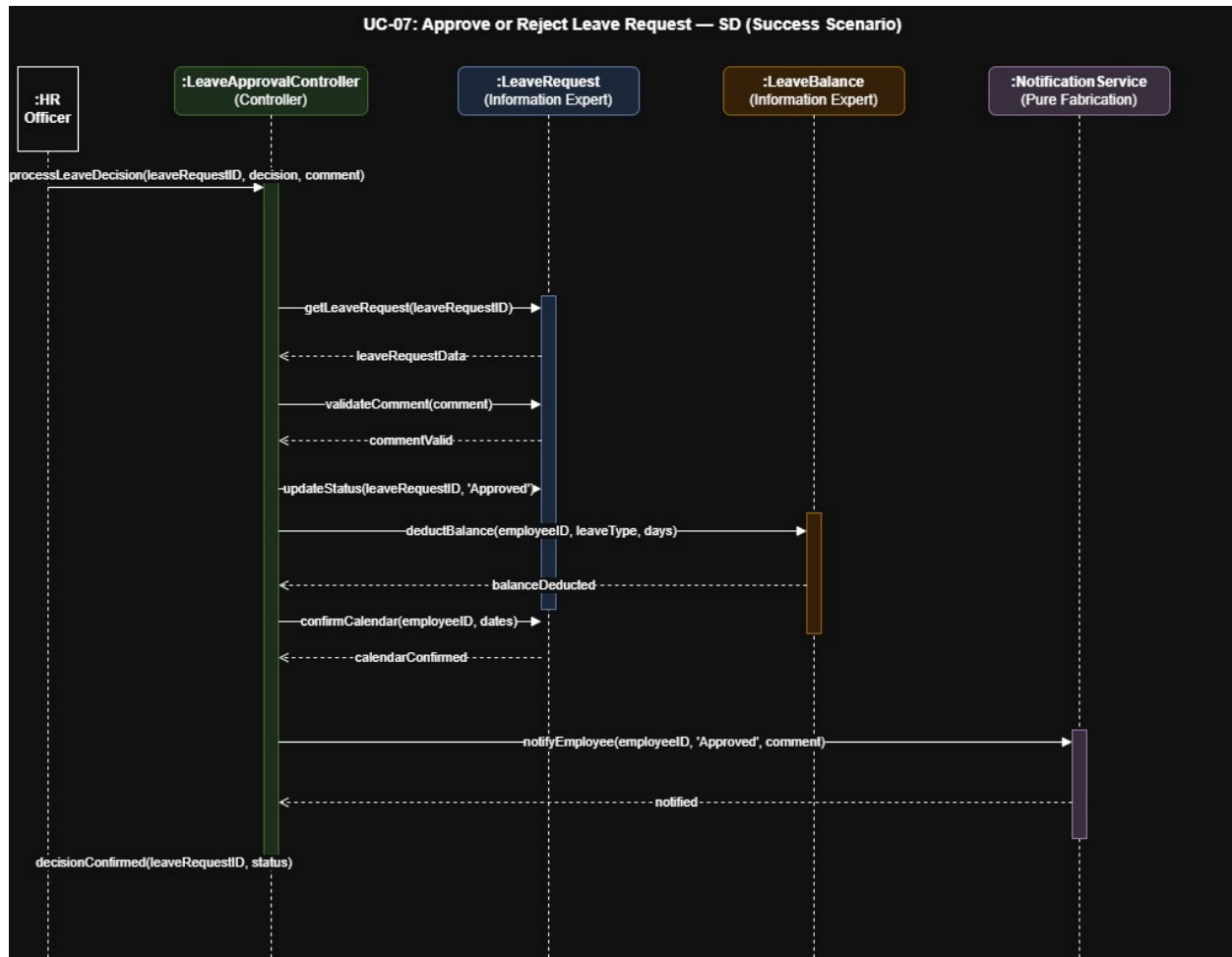


3.2 Leave and Attendance Management Module (UC-06 – UC-10)

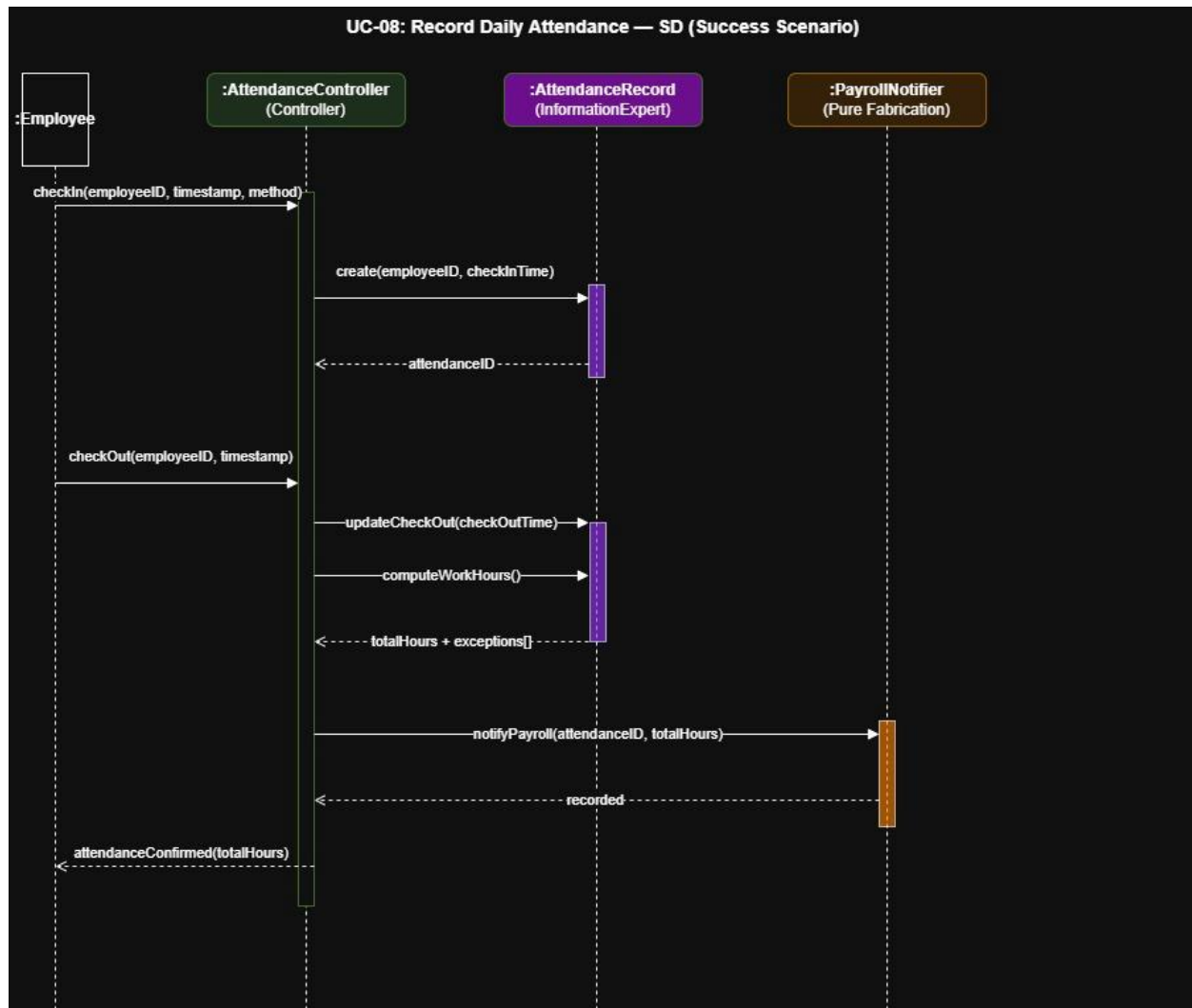
UC-06 – Sequence Diagrams



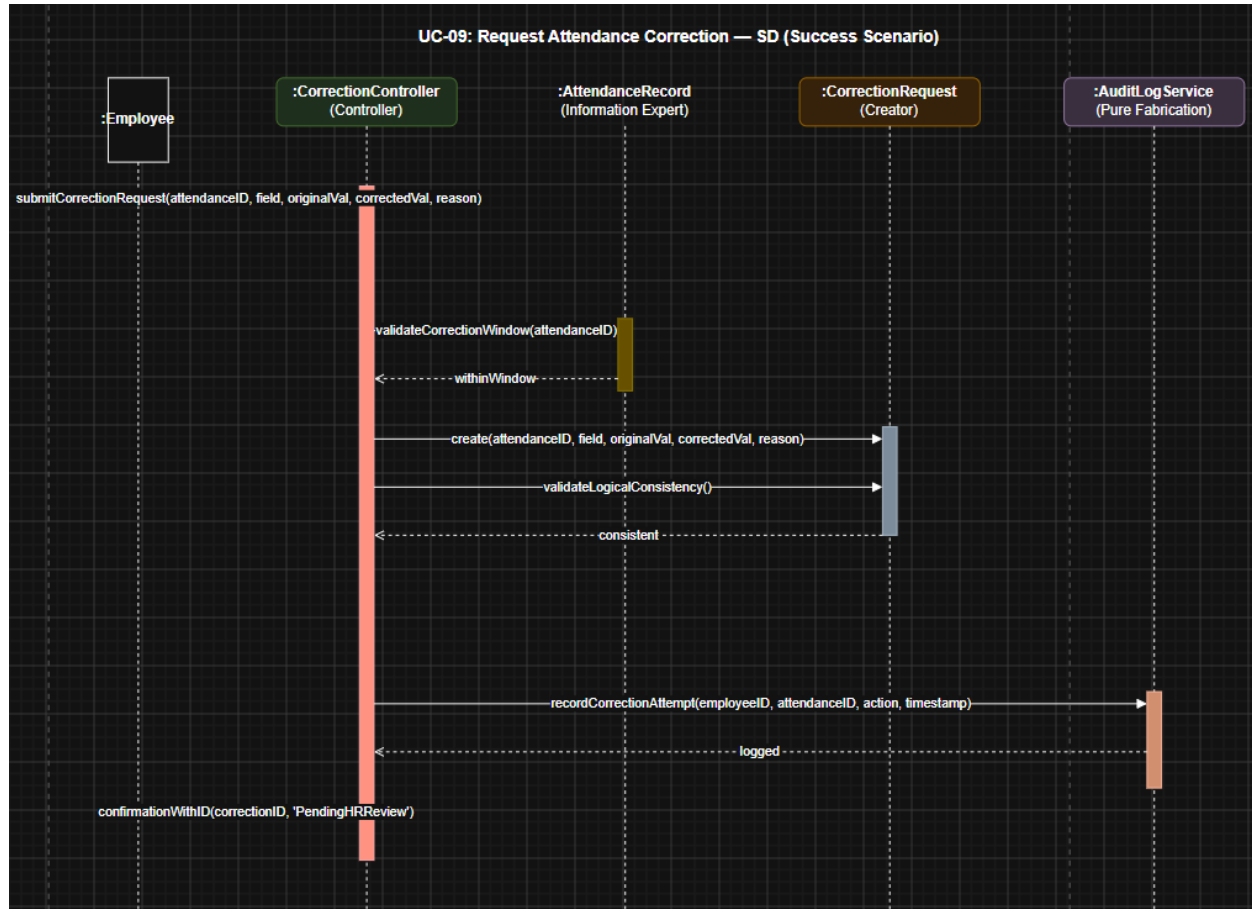
UC-07 – Sequence Diagrams



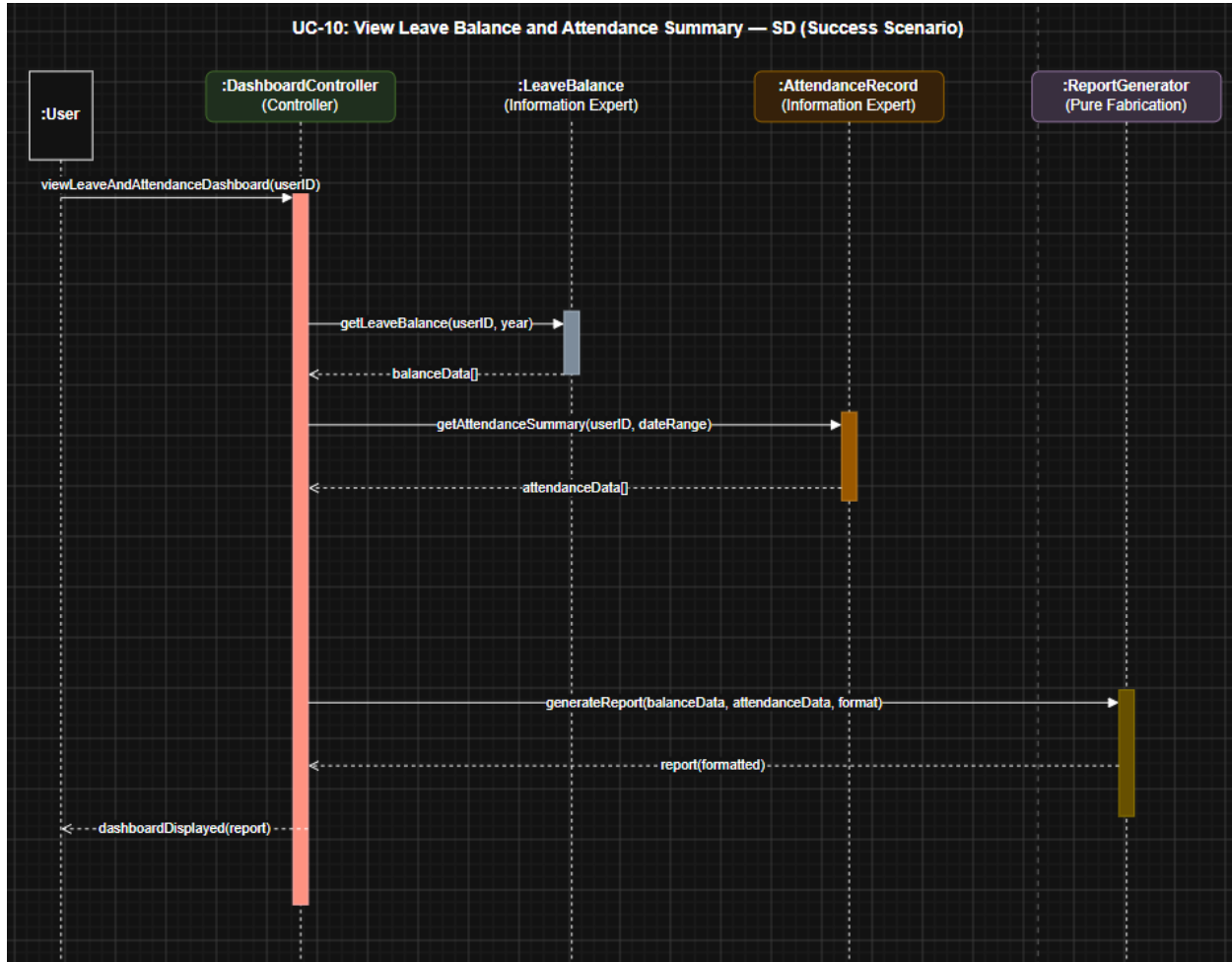
UC-08 – Sequence Diagrams



UC-09 – Sequence Diagrams

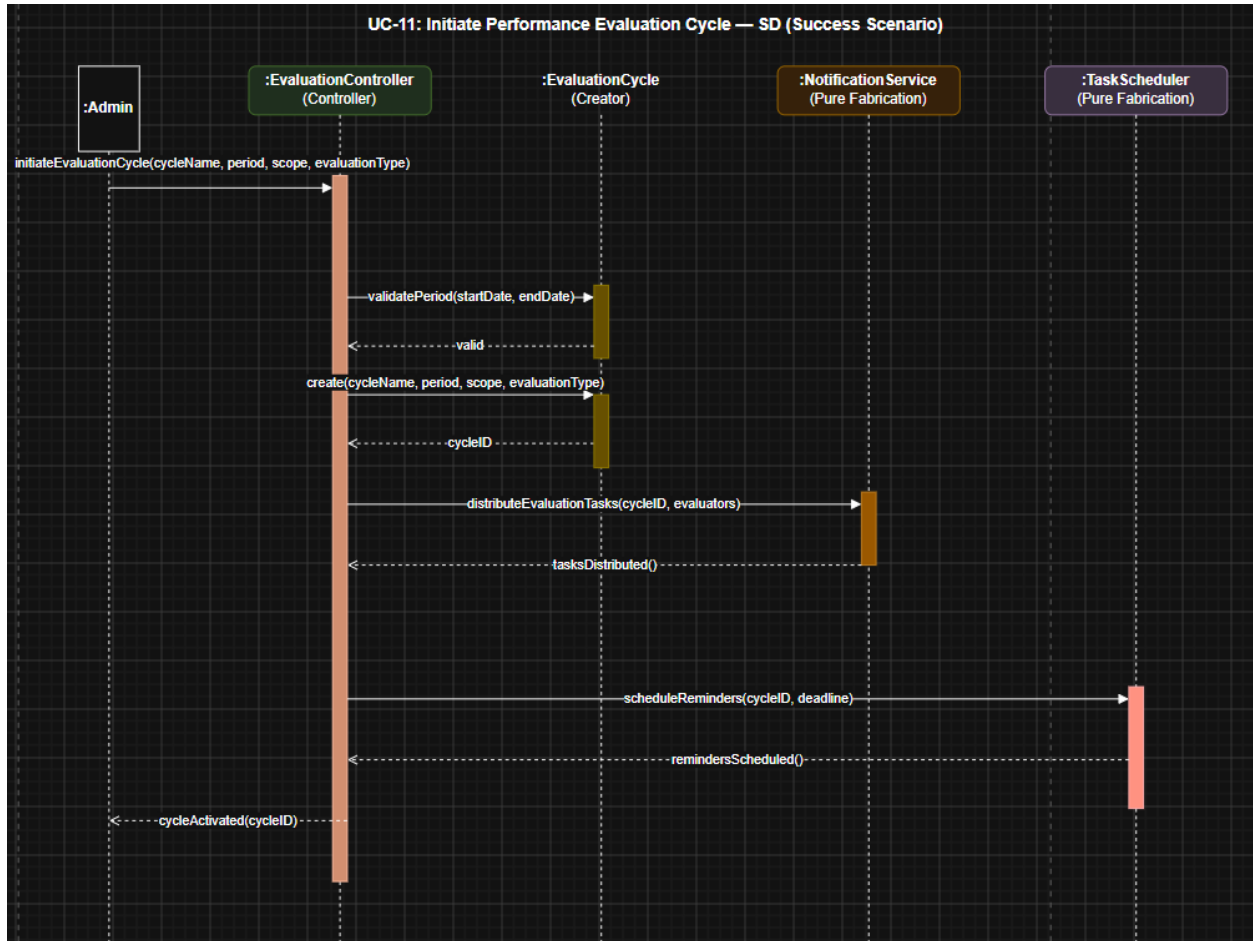


UC-10 – Sequence Diagrams

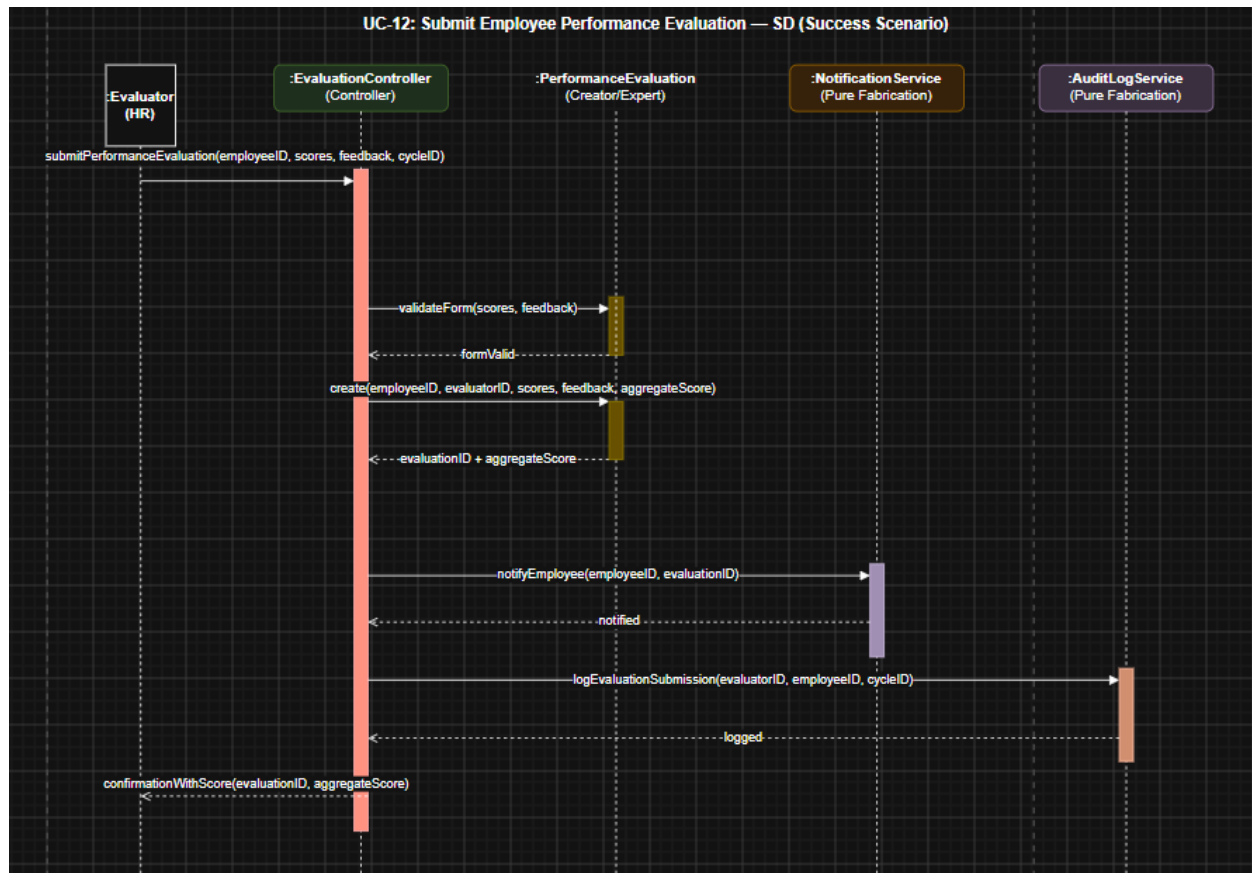


3.3 Performance and Compliance Management Module (UC-11 – UC-15)

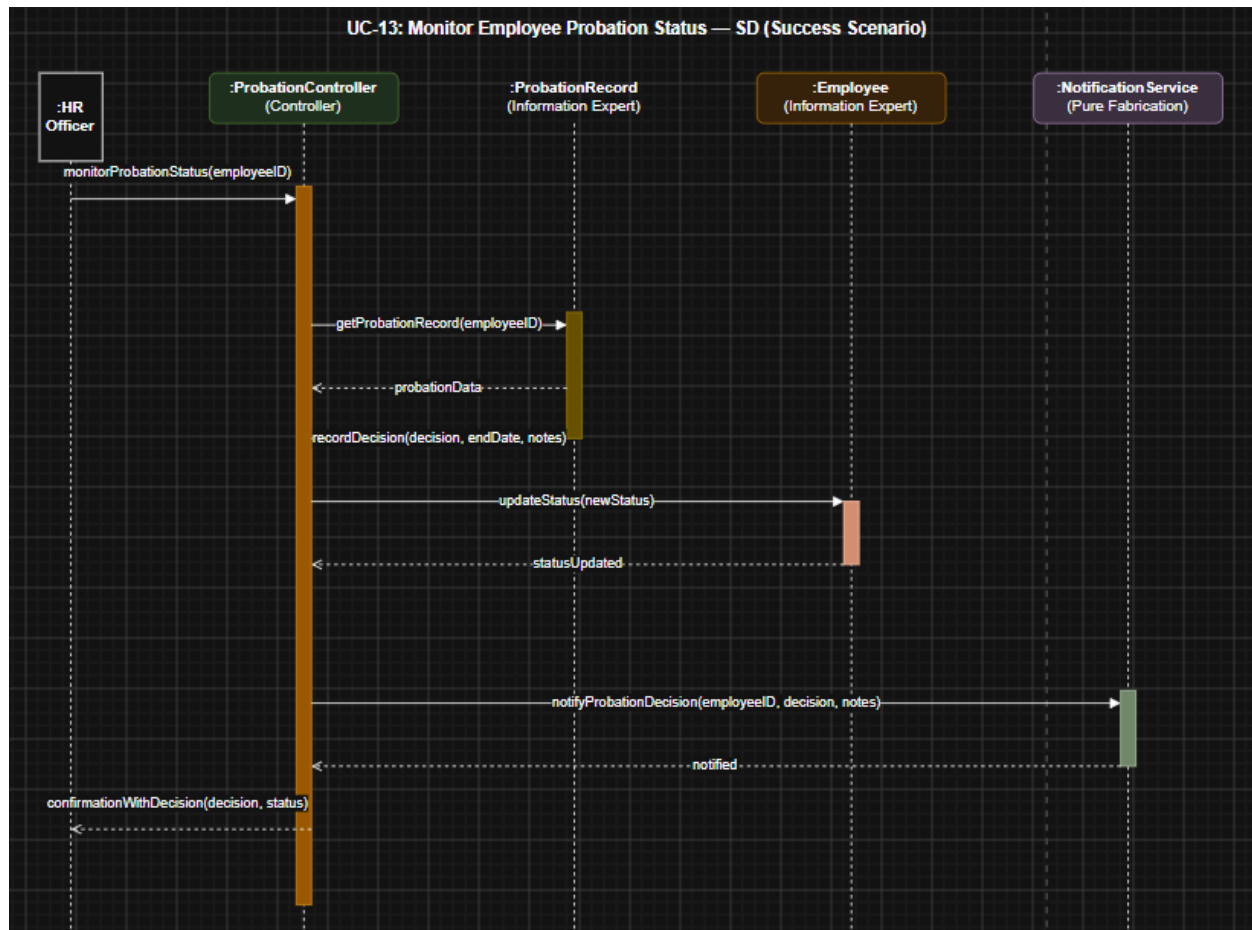
UC-11 – Sequence Diagrams



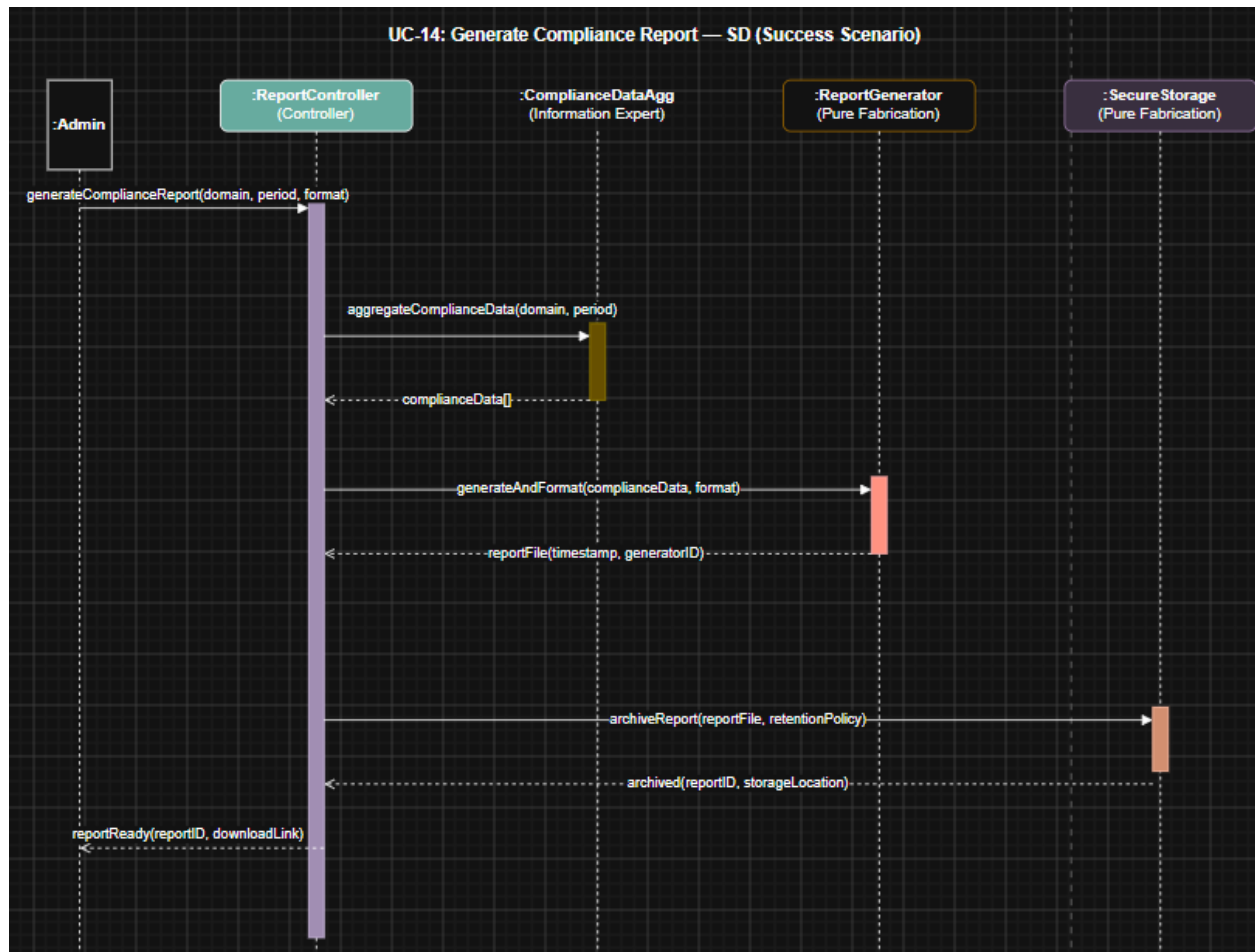
UC-12 – Sequence Diagrams



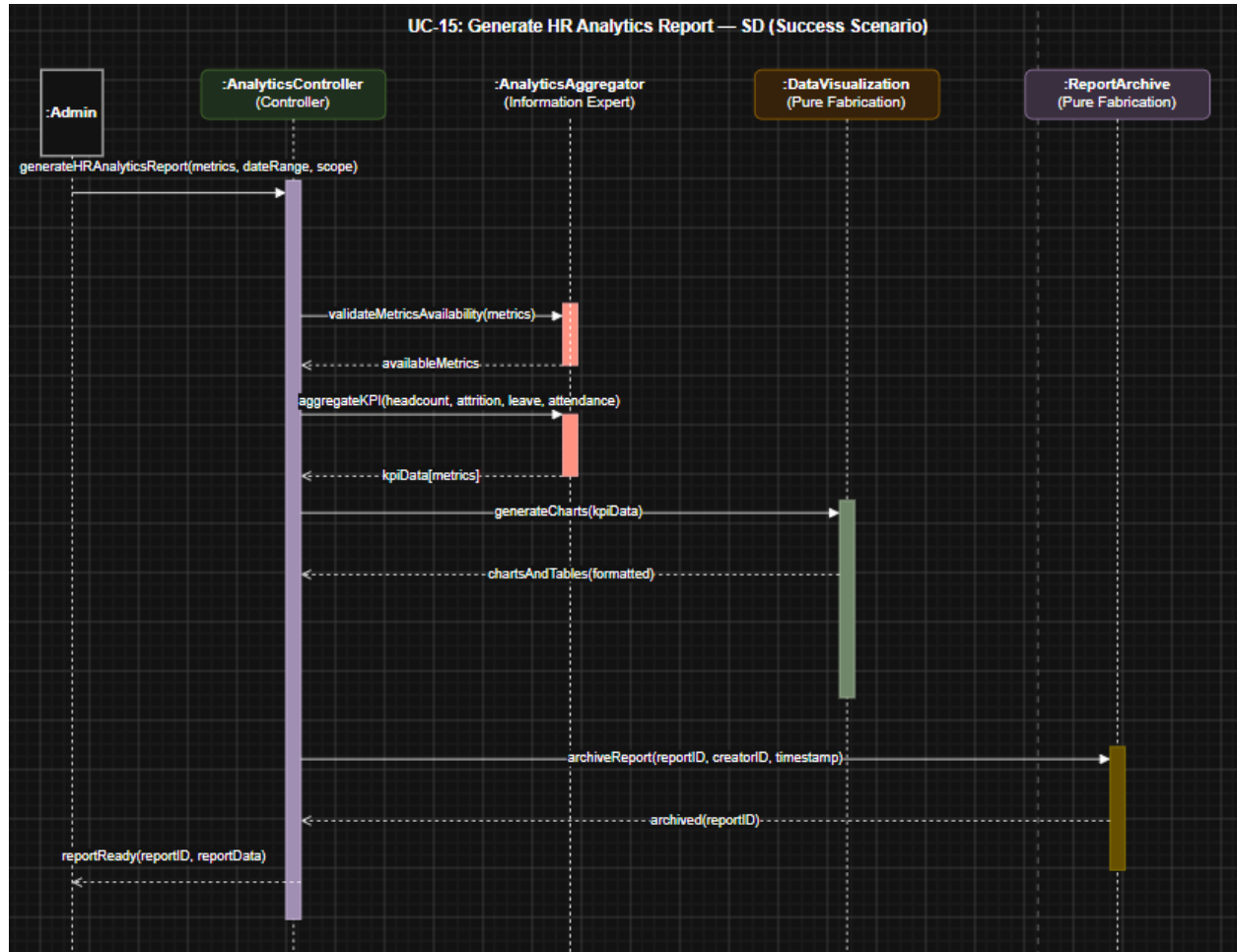
UC-13 – Sequence Diagrams



UC-14 – Sequence Diagrams



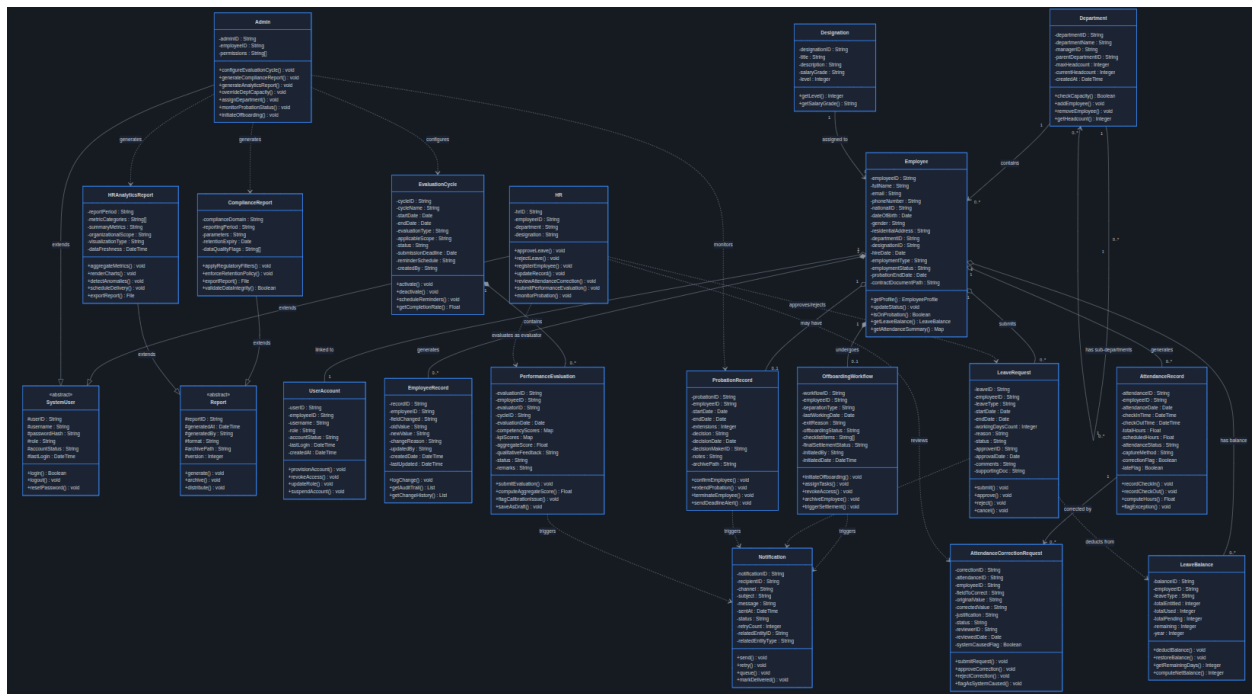
UC-15 – Sequence Diagrams



4. Class Diagram

The Class Diagram below integrates all operations assigned in the Sequence Diagrams and reflects the full domain model.

4.1 Class Diagram



5. Refined Domain Model

The Domain Model has been refined based on feedback to incorporate the OffboardingWorkflow entity, the LeaveBalance entity, and additional relationships identified during SSD and SD analysis. New attributes and associations added in this revision are highlighted in the diagram below.

